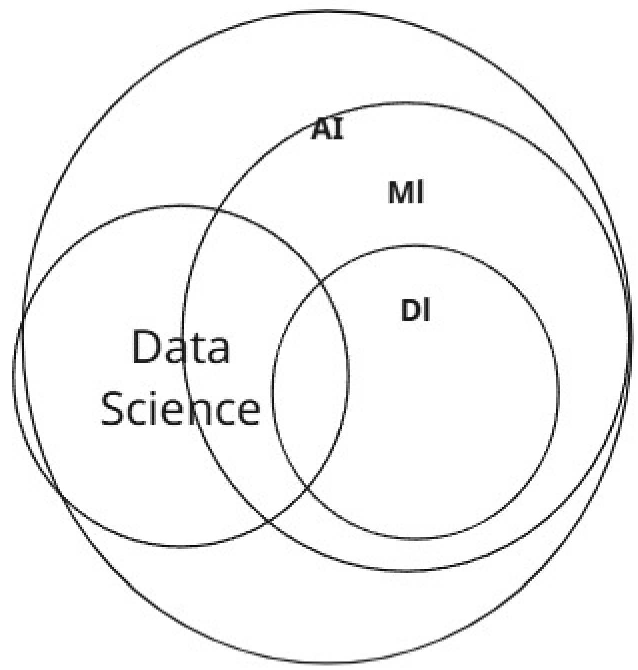
Machine Learning



Machine Learning

Machine Learning

^

^

Model

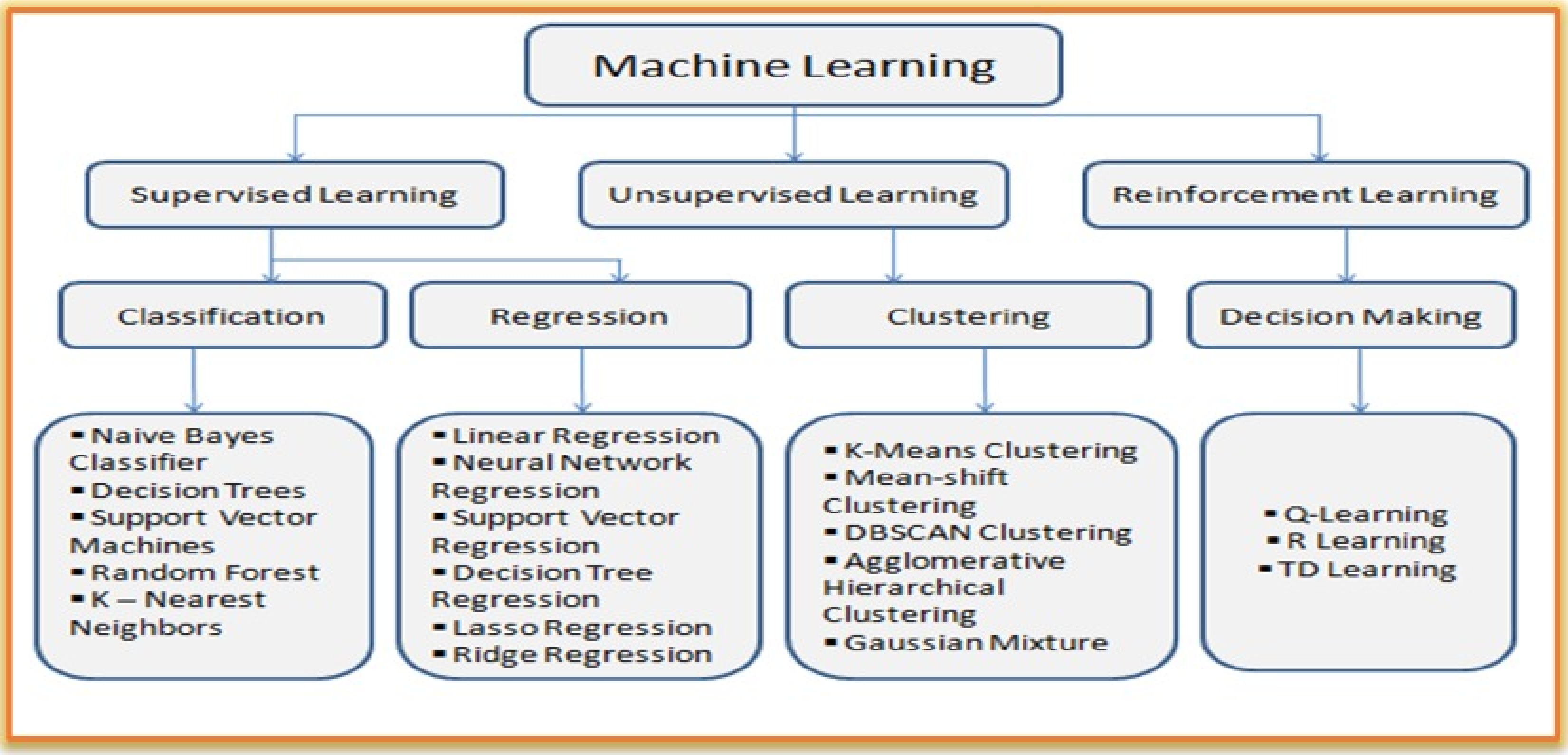
Types of Ml:-

1 Reinforestment learning >>

2 Supervised Machine learning>>> label data

3 Unsupervised Machine learning >>> unlabel data

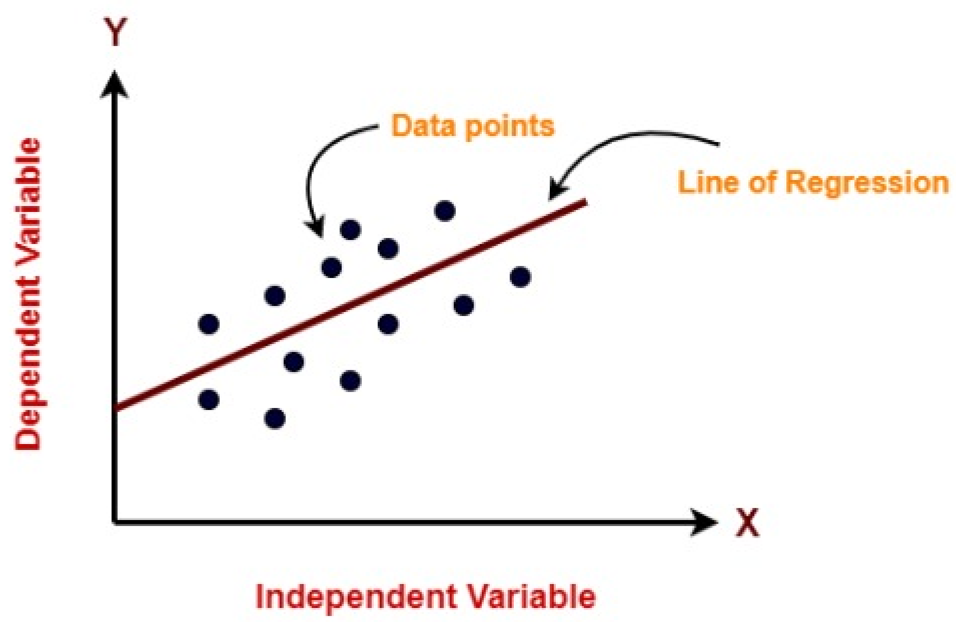
4 Semisupervised learning >>



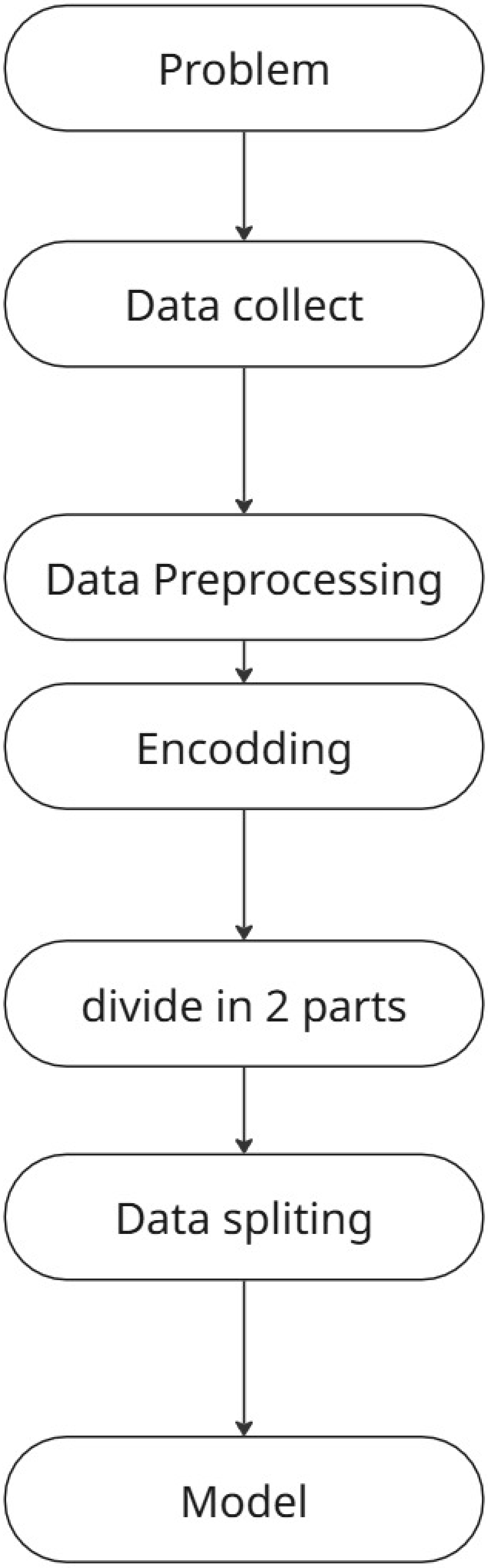
Supervised Machine learning>>> label data

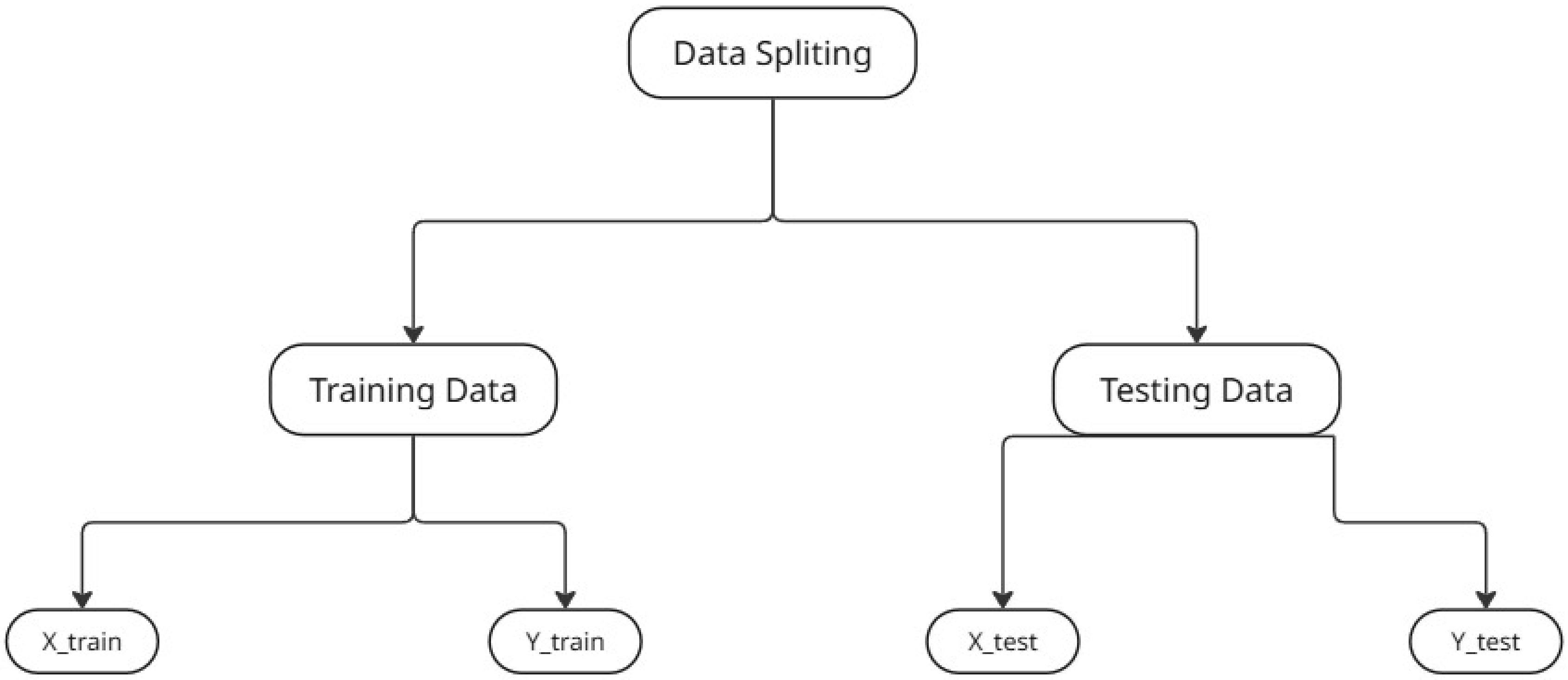
Linear Regression
strong the relationship is between two variables

$$y = mx + c$$



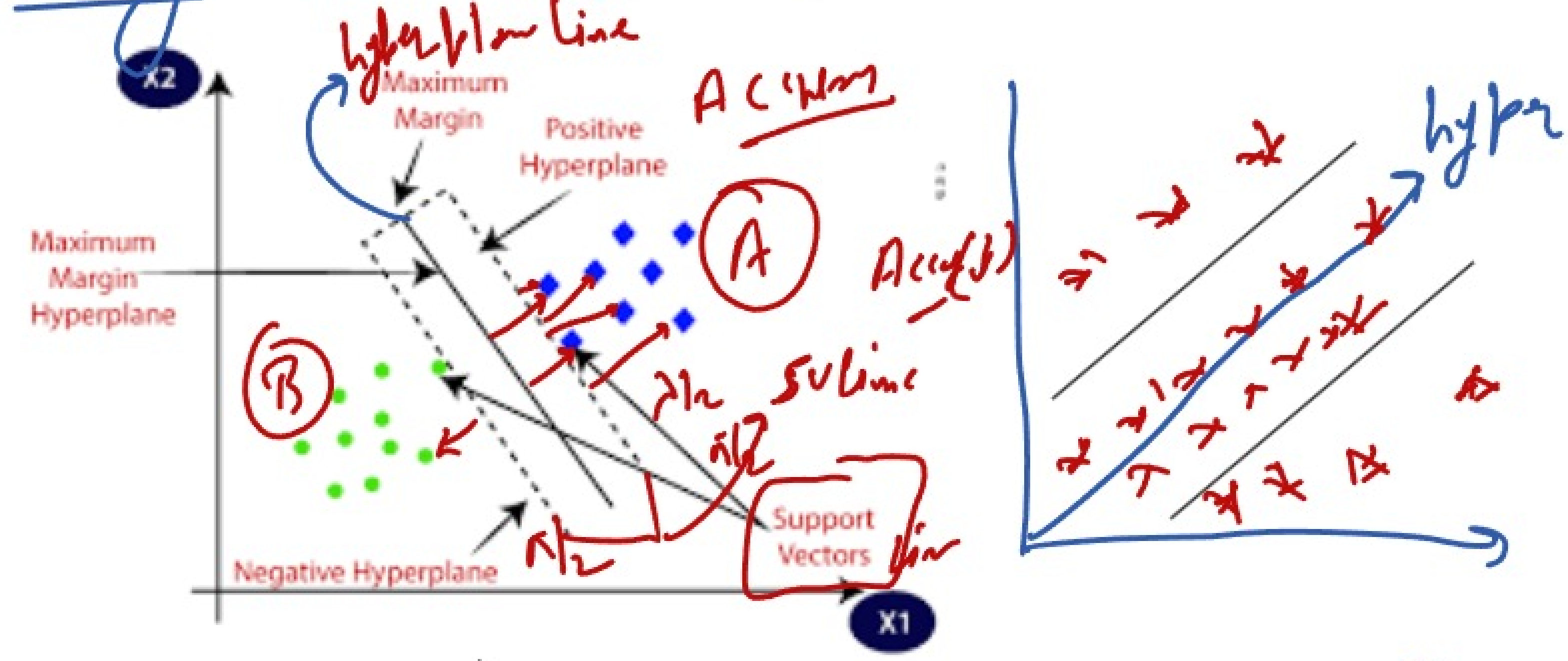
m= slope c = intercept
x = independent
y = dependent


Problem
Data collect
Data Preprocessing
Encodding
divide in 2 parts
Data spliting
Model

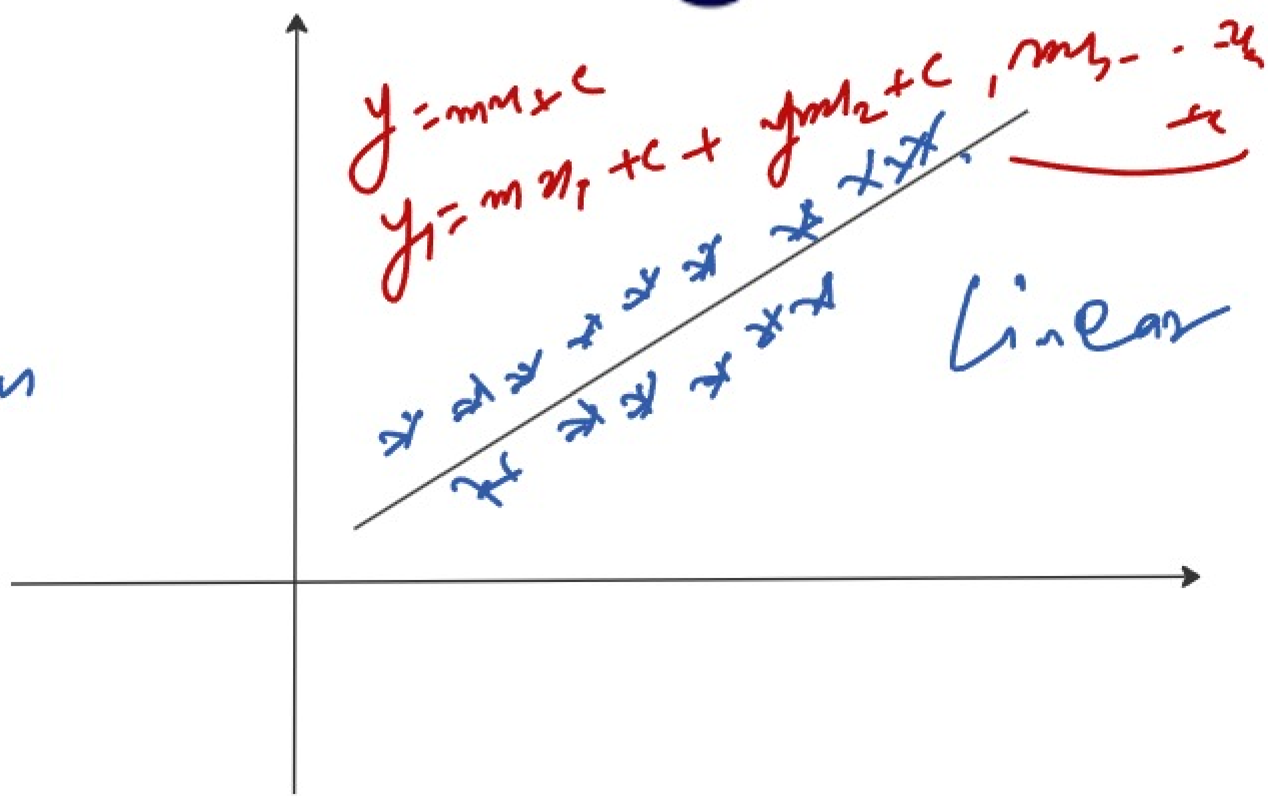


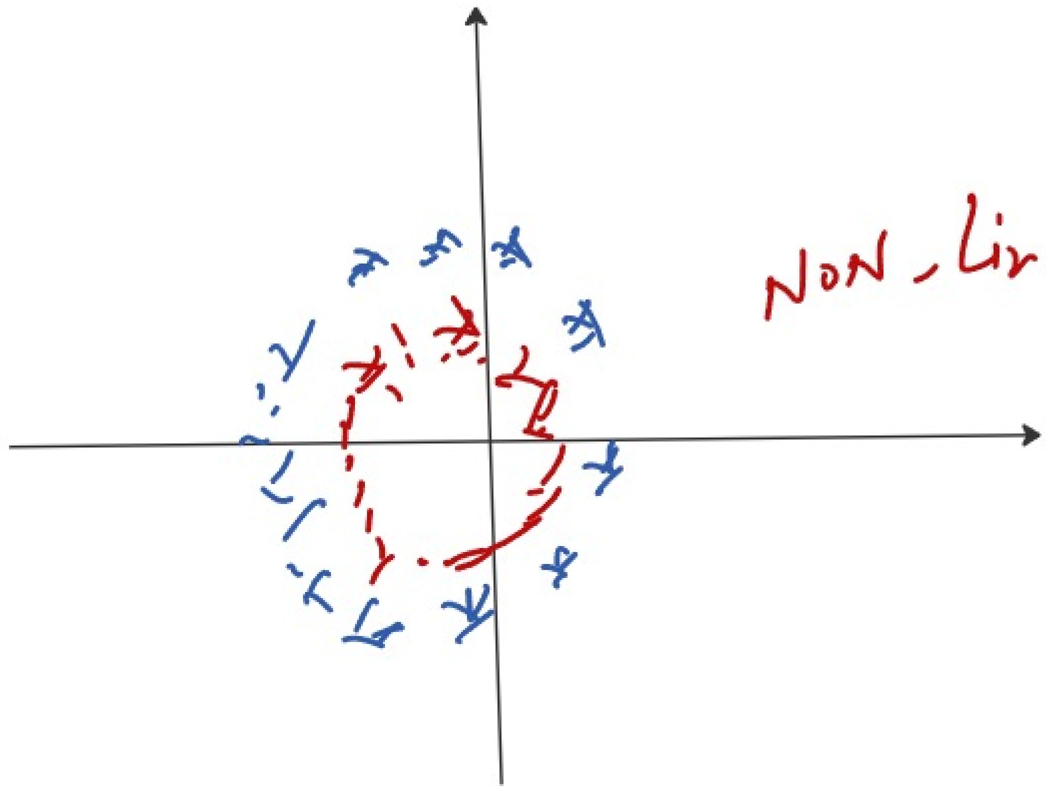
Support Vector Machine (SVM) Algorithm

supervised
classification & Regression
mostly used classification



types of SVM
Linear SVM
Non-Linear SVM

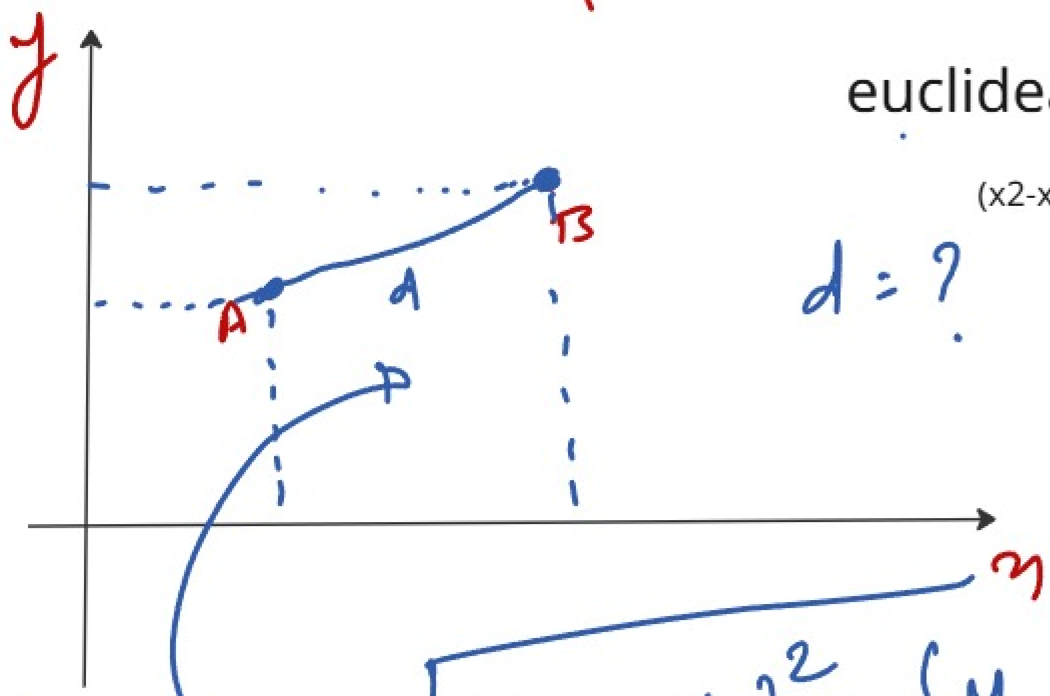




Non_linear data

k-nearest neighbors (KNN) algorithm

Sup
classification & Regression



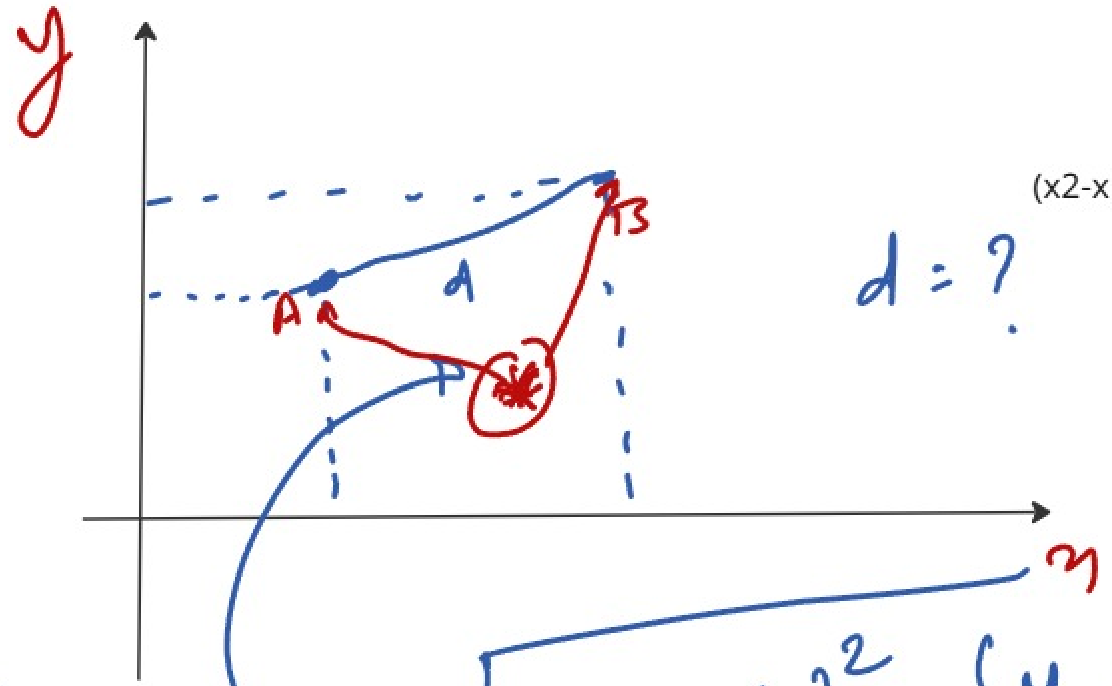
euclidean distance

(x2-x1)^2 + (y2-y1)^2 and upar root

$d = ?$

root (x2-x1)^2+(y2-y1)^2

Euclidean dis $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$



(x2-x1)^2 + (y2-y1)^2 and upar root

$d = ?$

root (x2-x1)^2+(y2-y1)^2

elidean dis $\rightarrow \sqrt{(x_2-x_1)^2+(y_2-y_1)}$

Ensembel learning

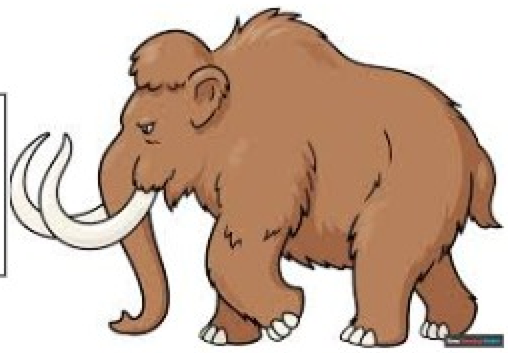
Mantri

King

$$\frac{23+23.5+24+24+18+20k \ldots m}{m}$$

2450kg

⇒ 2500kg
weight



23k
23.5k
24k
20k
18k

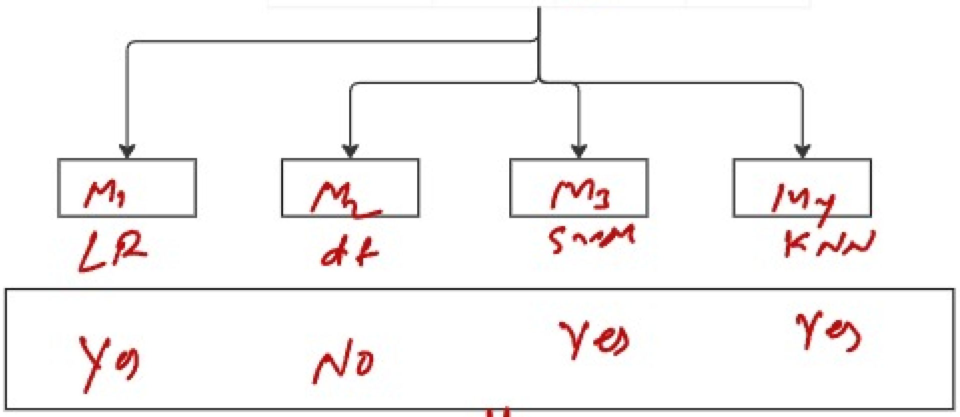
2400

Collage student data set

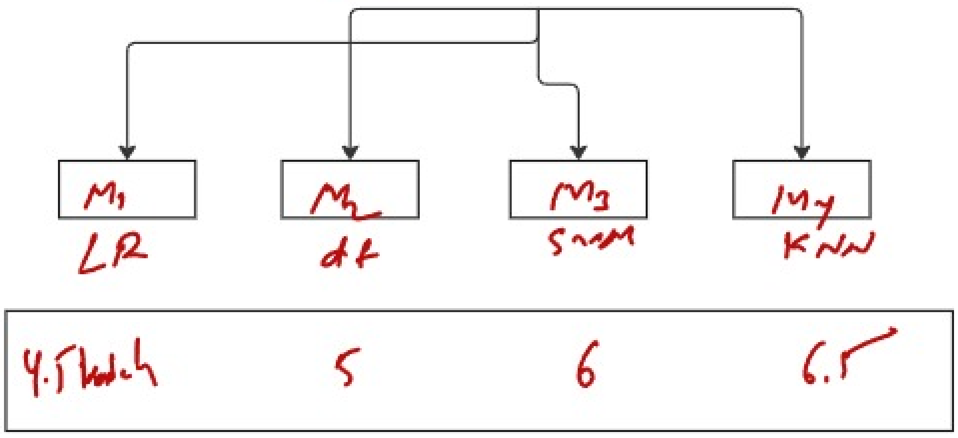
CGPA	IQ	Placement	
6.7	150	1	
8.7	150	0	
9.0	175	1	
8.0	130	1	
8.5	100	0	

College student data set

GPA	IQ	Placement	Salary
6.7	150	1 yes	5 lakh
8.7	150	0 No	0
9.0	175	1 yes	4.5
8.0	130	1 yes	6.5
8.5	120	No 0	7 lakh



Yes



$$\frac{4.5 + 5 + 6 + 6.5}{4} \Rightarrow \text{final}$$

Types of Ensemble learning => 4

(i) Voting
(ii) Stacking
(iii) Bagging
(iv) Boosting

Votting Ensembel

Algo - diff.

M_1	M_2	M_3	M_4
LR	df	SVM	KNN
Yes	No	Yes	Yes

⇓

(ii) Stacking Ensembel learning

M_1	M_2	M_3	M_4
LR	df	SVM	KNN
Yes	No	Yes	Yes

Algo	Pre
LR	Yes
df	No
SVM	Yes
KNN	Yes

Random forest

↓

final result

Bagging

Algo - Same

weak learnin

vertical

Boosting

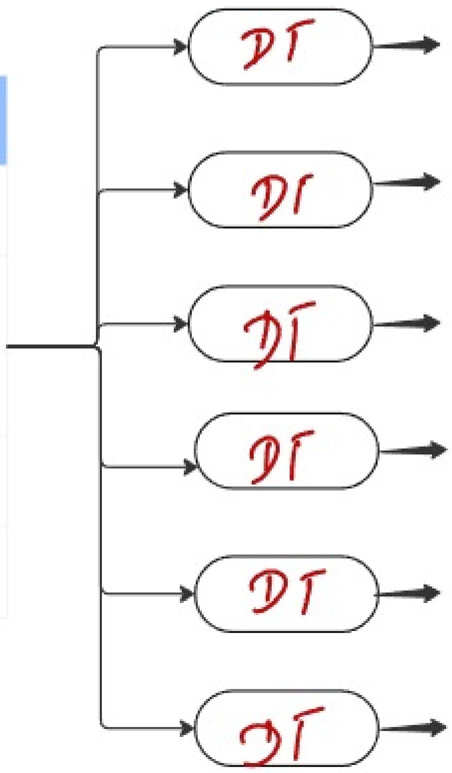
Algo same

weak

Horizontal

Bagging

cgpa	IQ	Placement	
6.7	150	1	
8.7	150	0	
9.0	175	1	
8.0	130	1	
8.5	110	0	

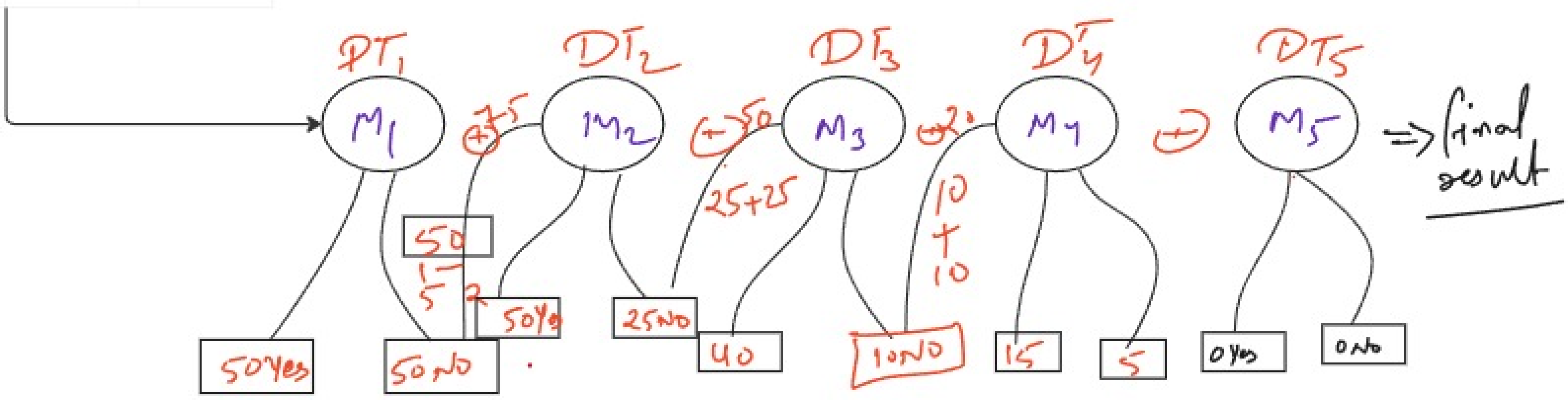


(i) Random forest

(ii) Boosting

(i) Adaboost
(ii) XGboost
(iii) Gradient boosting Algo.

cgpa	IQ	Placement	
6.7	150	1	
8.7	150	0	
9.0	175	1	
8.0	130	1	
8.5	110	0	



Random forest

sup

DT1 DT2 DT3 DT4 DT DTn

classification & Regression

overfitting Problem

weak lear. train

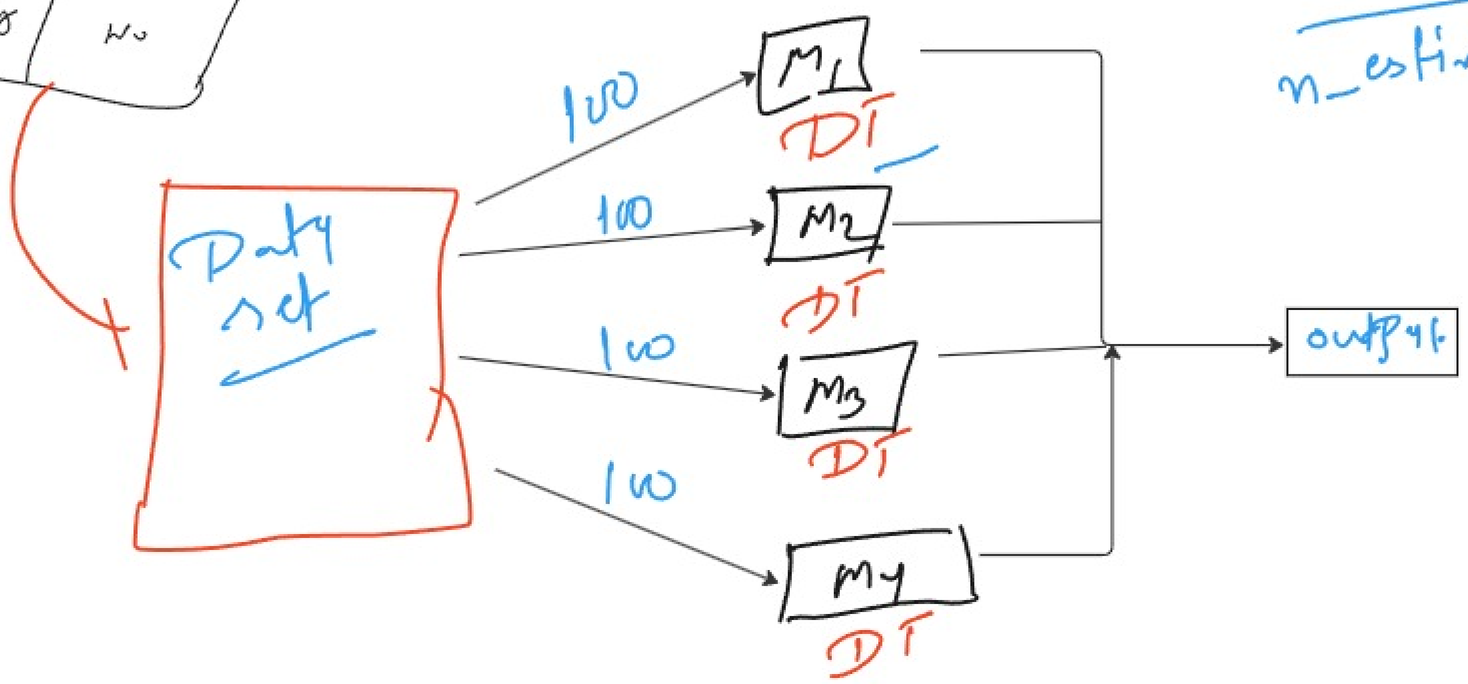
student

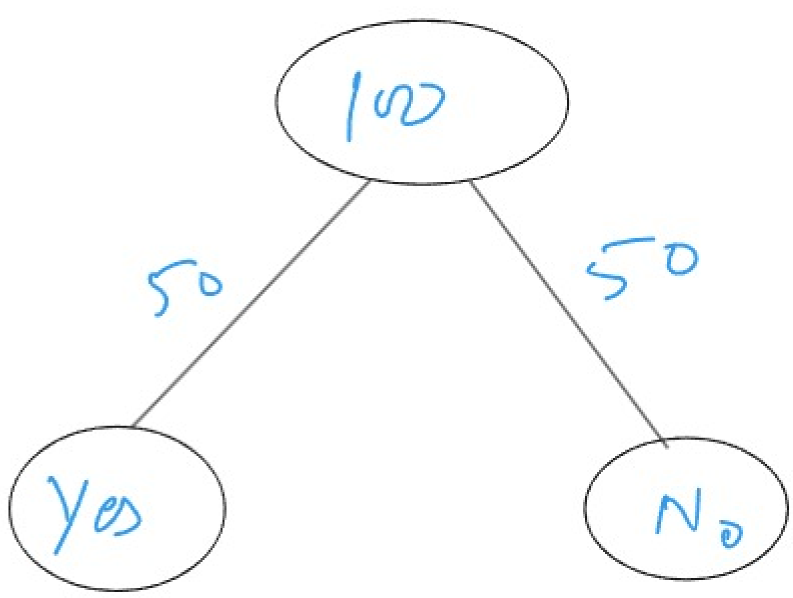
10

CGPA	IQ	Placement
8.5	75	yes
9.5	85	no
7.5	73	yes
10	78	no

#

Estimator = dt
n_estimator = 100
100
50



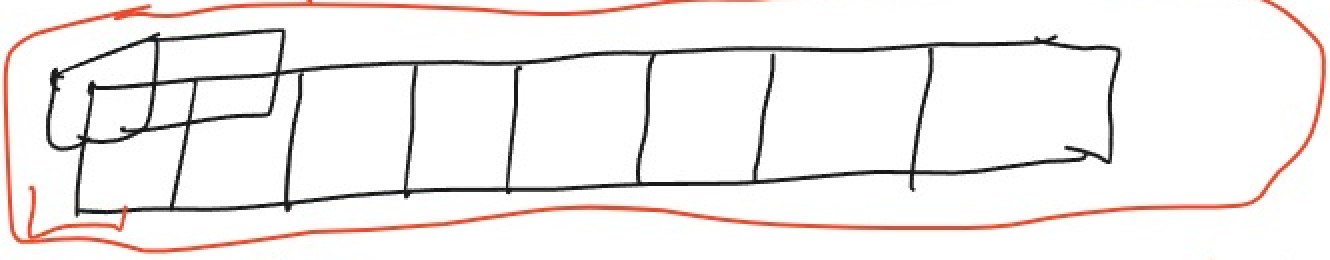


1. Entropy
2. gini
3. info gain

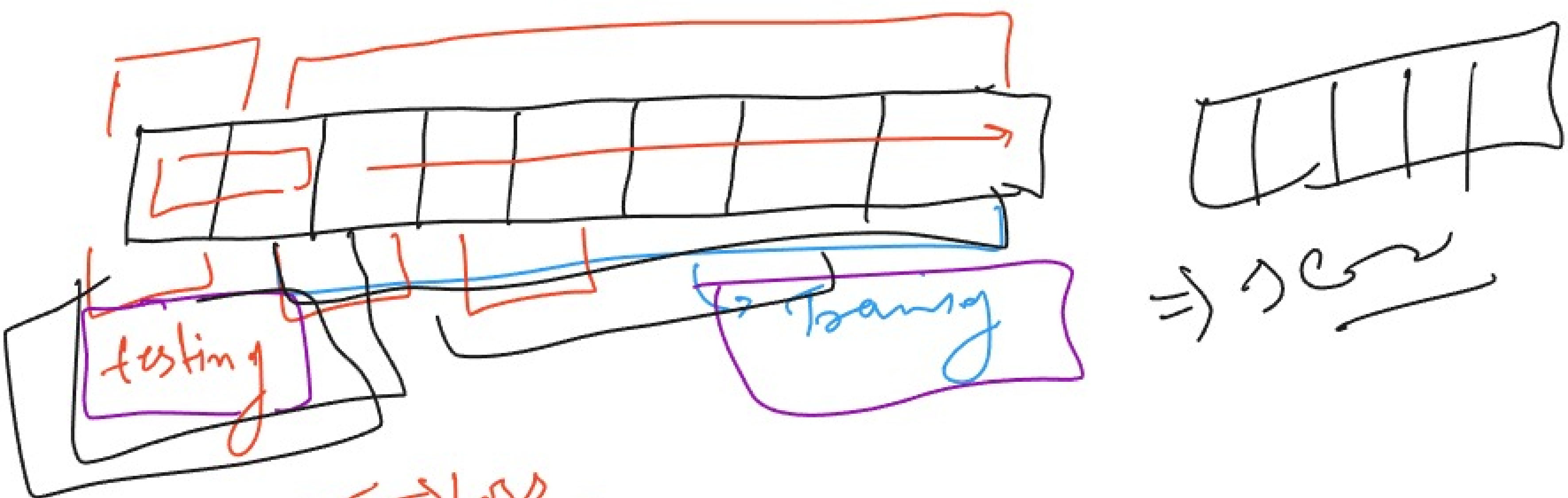
(I) oob score (out of Bag score)

(II) oob error (out of Bag error)

(I) Oob score (out of Bag score)



K-fold validation CV ⇒ cross validation



(II) Oob error ⇒ loss

1 − oob score

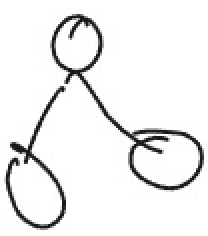
validation error

Tree

hyperparameter

(I) n estimator ⇒ define no. of DT
By default = 100

(II) criterion ⇒ entropy, gini, info. gain
score



(III) max depth = By default = 3.

(IV) Min_sample_split = By default = 2

(V) Min sample_leaf =

(VI) Oob_score By default = false

True

from sklearn.ensemble import Randomforestclassifier
Randomforestregressor

obj = Randomforestcls(n_estimator = 150, max depth = 3,
min_sample_node = 1, oob_score = True)
false

—120 => 100
100
105 —106

obj = Randcls (n)
obj.fit (x

obj.get_Param_

✅ OOB (Out-of-Bag) Score

- Measures how well the model performs on **OOB samples** (data not used during training for each tree).
- It's like an internal **validation accuracy**.
- Expressed as a **score** (often accuracy for classification, or R^2 for regression).
- **Higher is better**: close to 1.0 = good model.

❌ OOB Error

- The **complement** of OOB score.
- Represents the **average prediction error** on OOB samples.
- Often expressed as a **percentage**.
- **Lower is better**: close to 0 = good model

Task >>>>

projects

Bike price

ipynb file
csv
model >>>> Ramdomforest regressore

laptop price

ipynb file
csv
model >>>> Ramdomforest regressore

ham and spam

ipynb file
csv
model >>>> Ramdomforest regressore

disease

ipynb file
csv
model >>>> Ramdomforest regressore

Naive

feature indepdent algo

Bayes theram

$$P(B/A) = \frac{P(B) * P(B/A)}{P(A)}$$

outlook	TM	Play for	
Sunny	H	No	
Sunny	H	No	
over cast	H	Yes	
Rain	H	Yes	
Rain	M	Yes	
Rain	C	No	
over cast	C	Yes	
Sunny	M	No	
Sunny	C	Yes	
Rain	M	Yes	
Sunny	M	Yes	
over cast	M	Yes	
over cast	H	Yes	
Rain	M	No	

$P(yes) = 9/14$

$P(no) = 5/14$

outlook	freq(yes)	freq(no)	Prob(yes)	Prob(no)
Sunny	2	3	2/9	3/5
overcast	4	0	4/9	0/5
Rain	3	2	3/9	2/5
Sum →	9	5		

if [Sunny, Hot]

P(Yes)

P(No)

$$P(Yes/[Sunny, H]) = \frac{P(Yes) \times P(Sunny/Yes) \times P(Hot/Yes)}{P(Sunny) \times P(Hot)}$$

$$P(No/[S, H]) = \frac{P(No) \times P(Sunny/No) \times P(Hot/No)}{P(Sunny) \times P(Hot)}$$

$$\boxed{\begin{array}{l} P(yes) = 0.75 \\ P(no) = 0.43 \end{array}}$$

for Yes $\dfrac{0.75}{0.75 + 0.43} = 0.62$

for no $= \dfrac{0.43}{0.75 + 0.43} = 0.36$

Print

(yes > no)

Yes

<u>Classification</u>

Types of Bayes

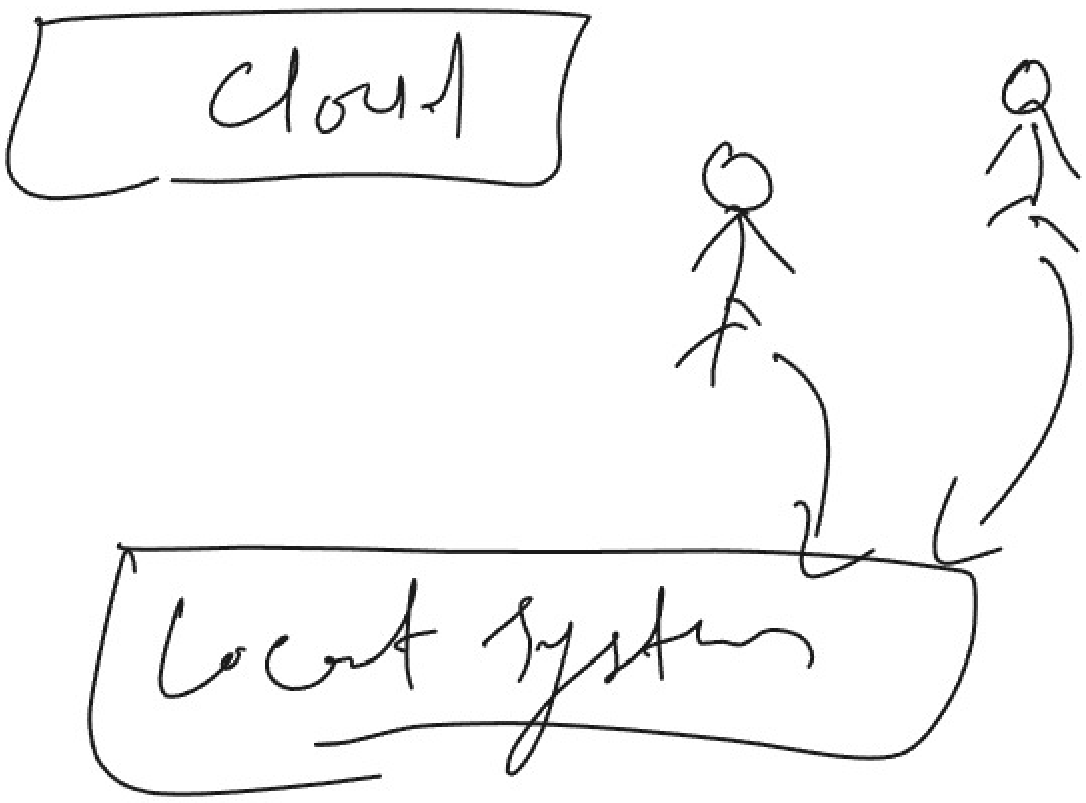
1. Gaussian Naive Bayes
2. Multinomial Naive Bayes
3. Bernoulli Naive Bayes

Yc Noor

from sklearn.naive_bayes import G, M, B

Aws

Cloud => बादल



On

Linode

Im

Ret

Kaggle

Unsupervised learning

Clustering

(I) Kmeans & Kmeans++

(II) DBSCAN

(III) Hierarchical clustering

association

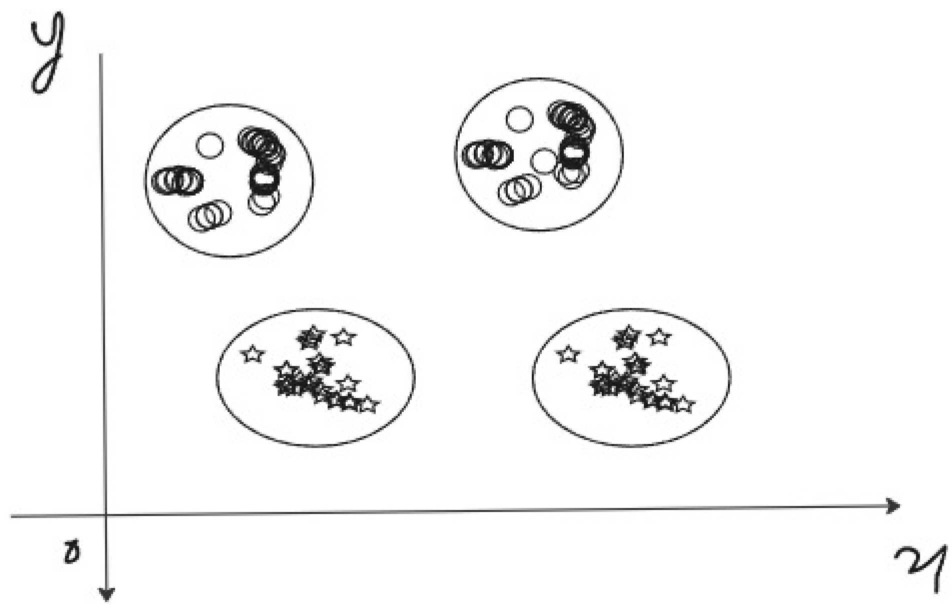
MBA

(II) Apriori

(I) Dimension Position

(I) PCA

Clustering ⇒



Kmeans & Kmeans++

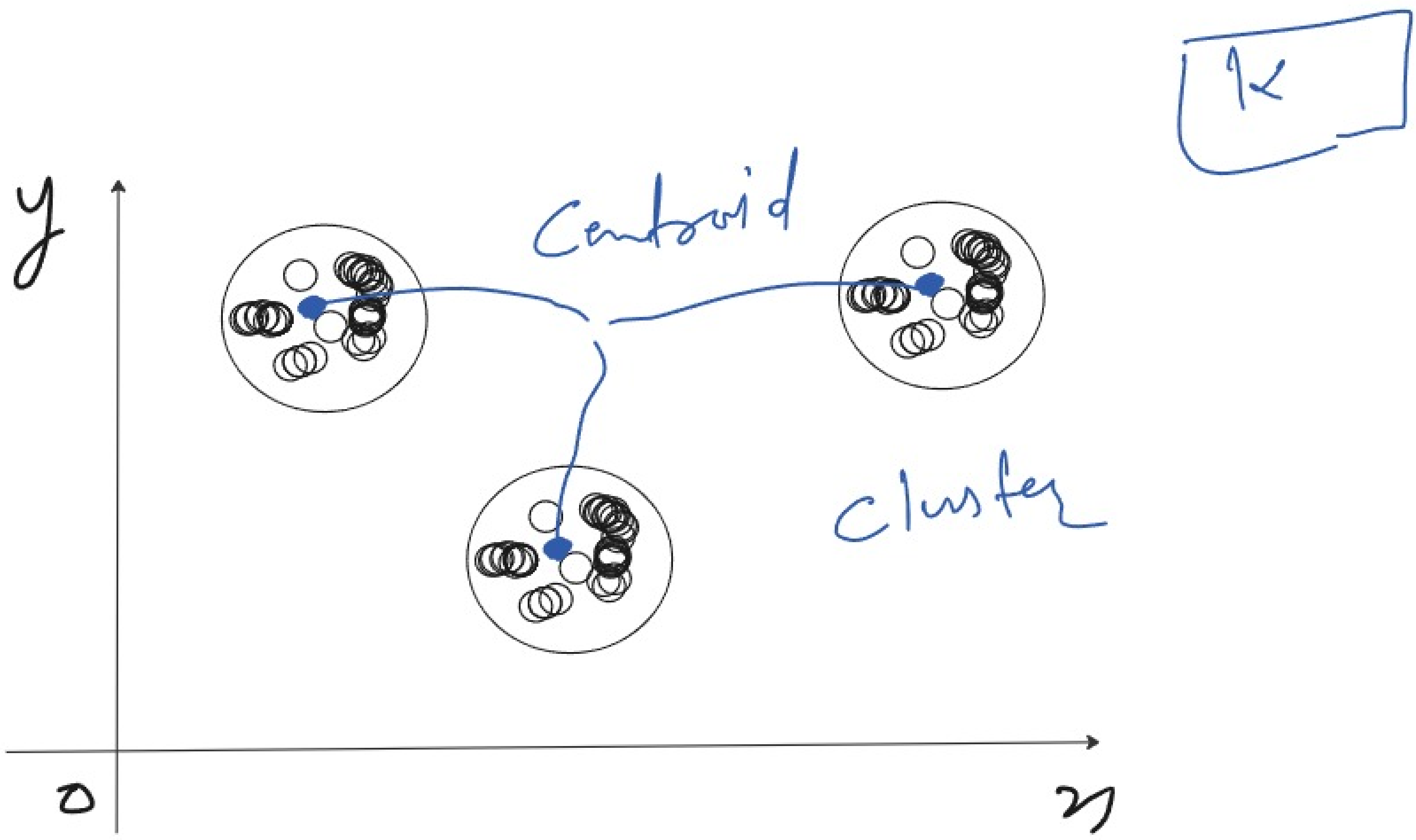
Unsupervised learning

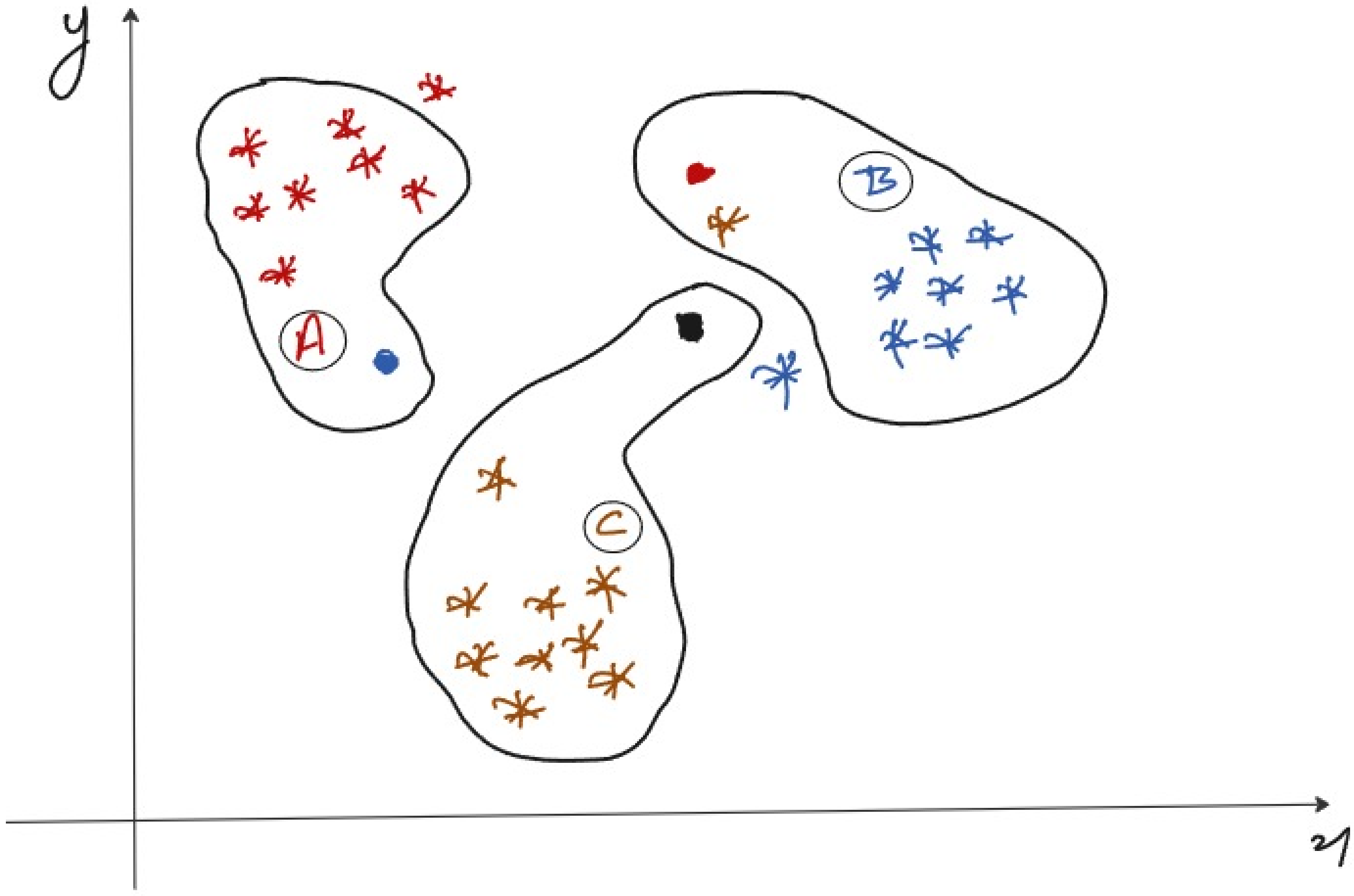
that is used to solve the clustering Problem

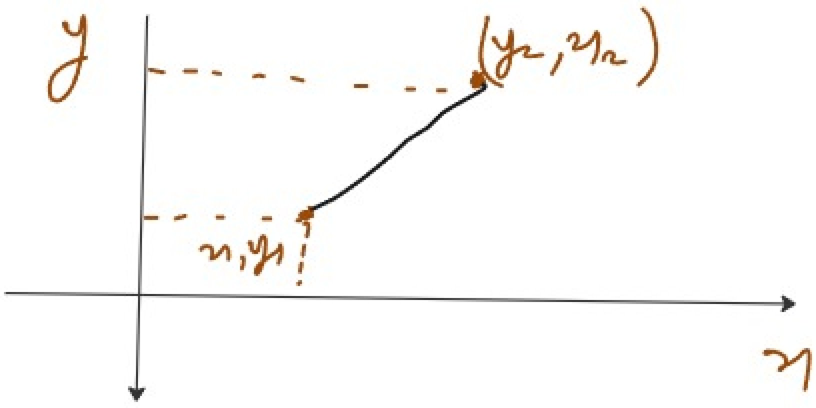
unlabeled

Kmeans

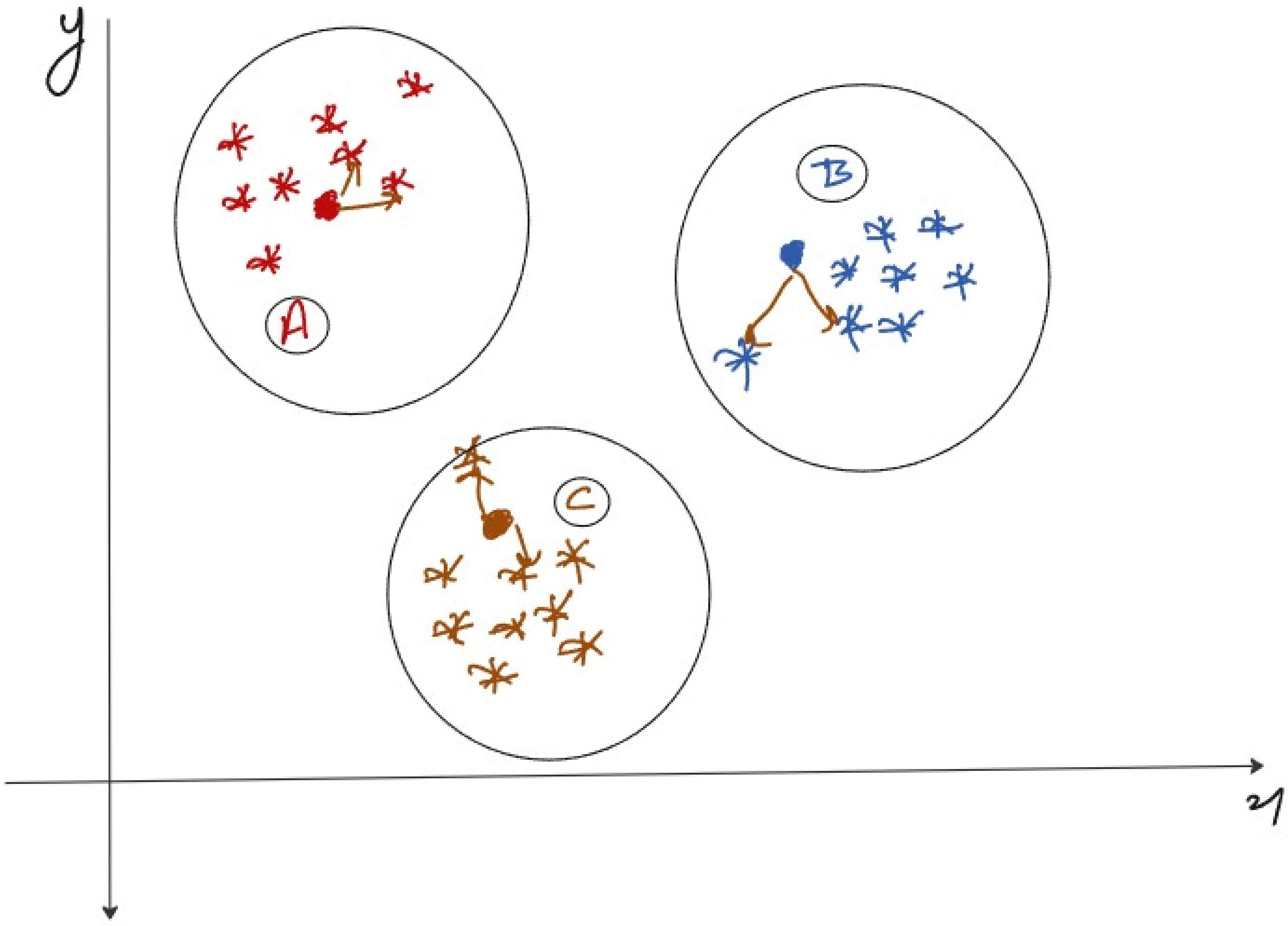
K = 1





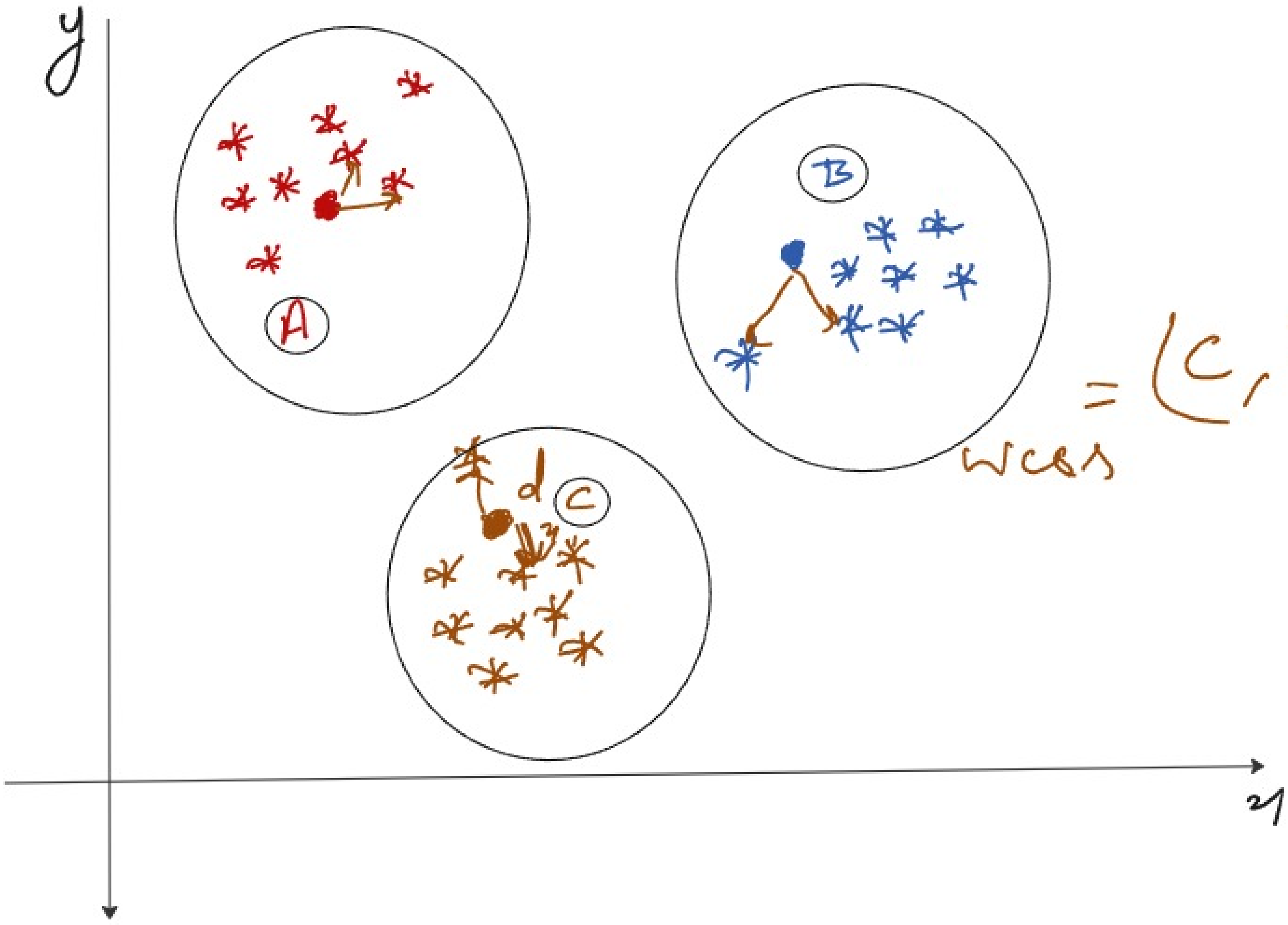


$$\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$



K = ?

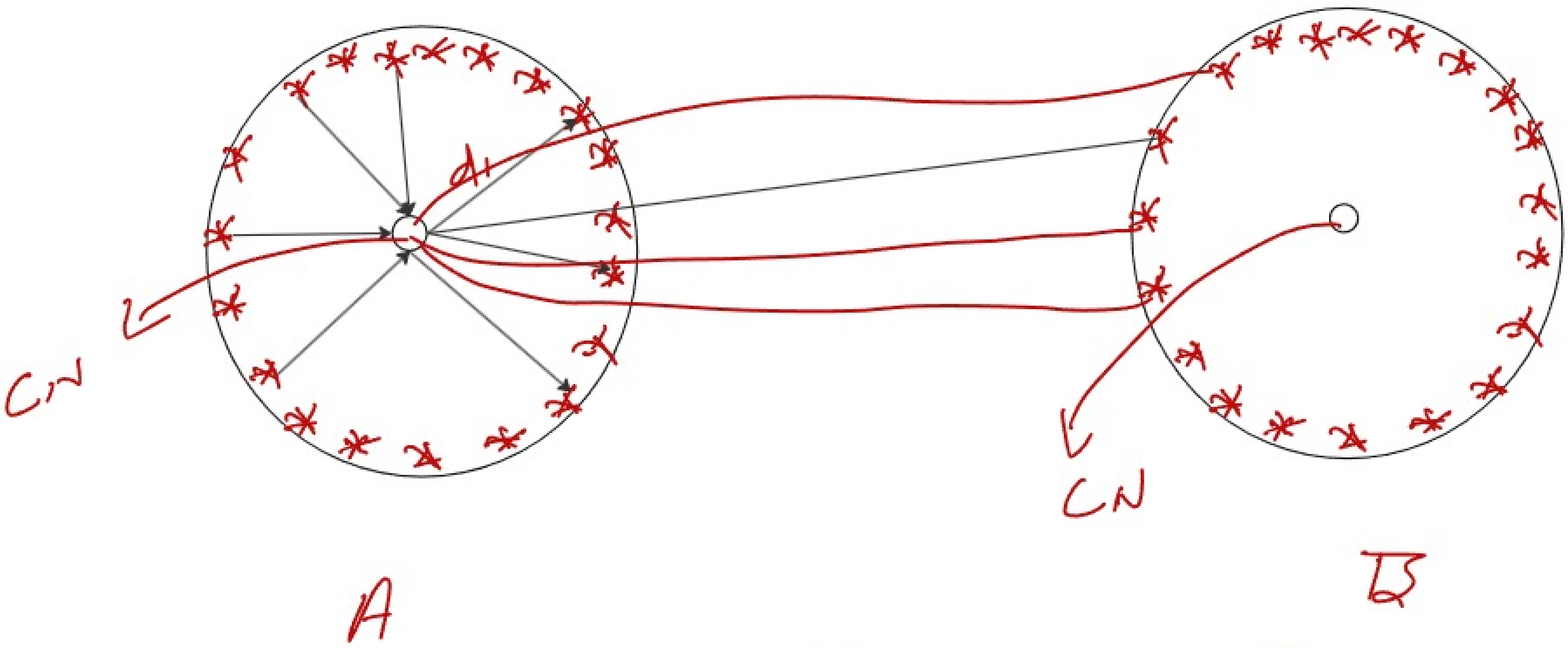
WCSS ⇒ within cluster sum of square



$$WCSS = (c, x_i)^2$$

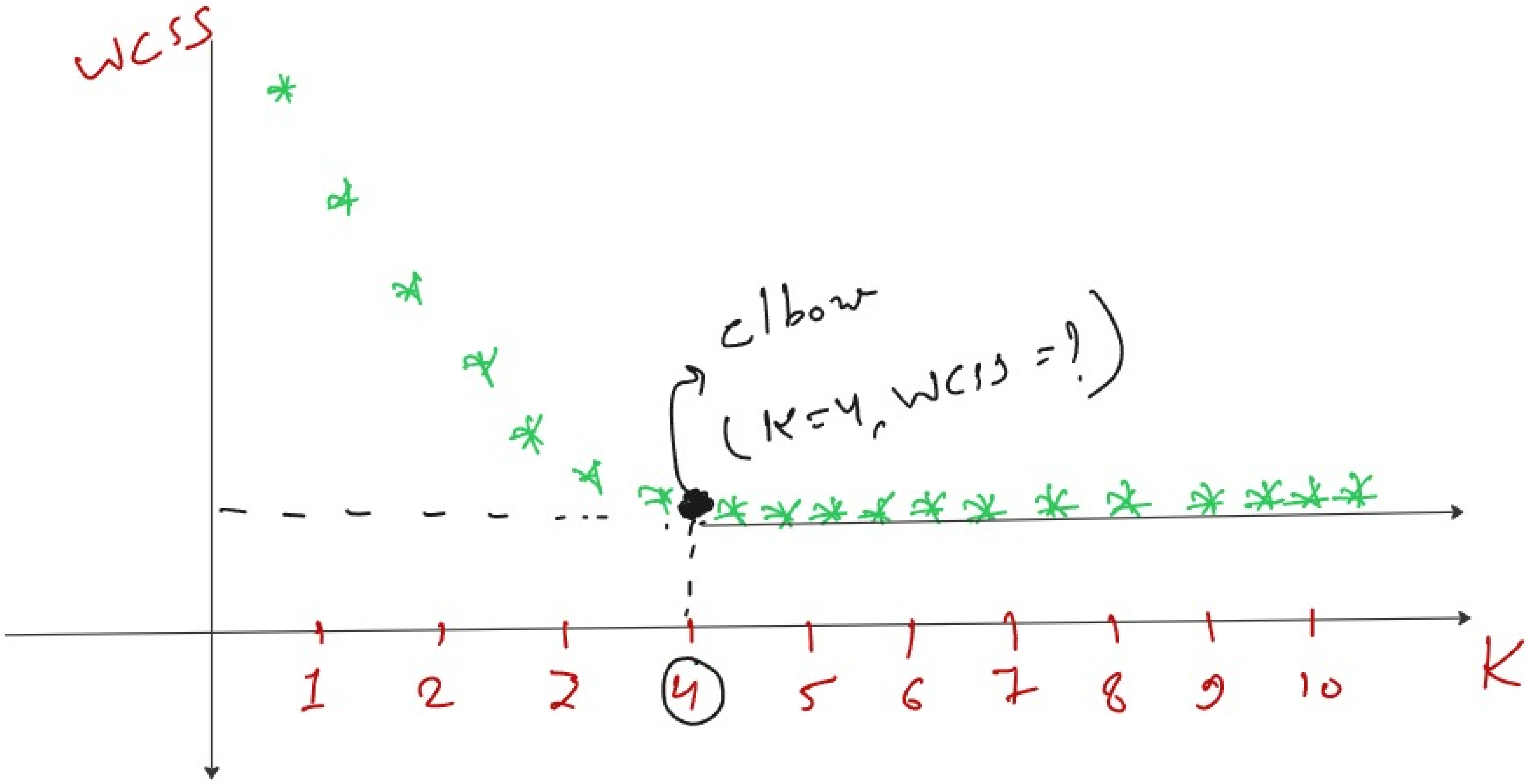
(i) intra cluster distance

(ii) inter cluster distance



$$WCSS = \sum_{i=1}^{n} d(c, x_i)^2 \text{ or } \underline{\text{inhertia}}$$

Elbow method

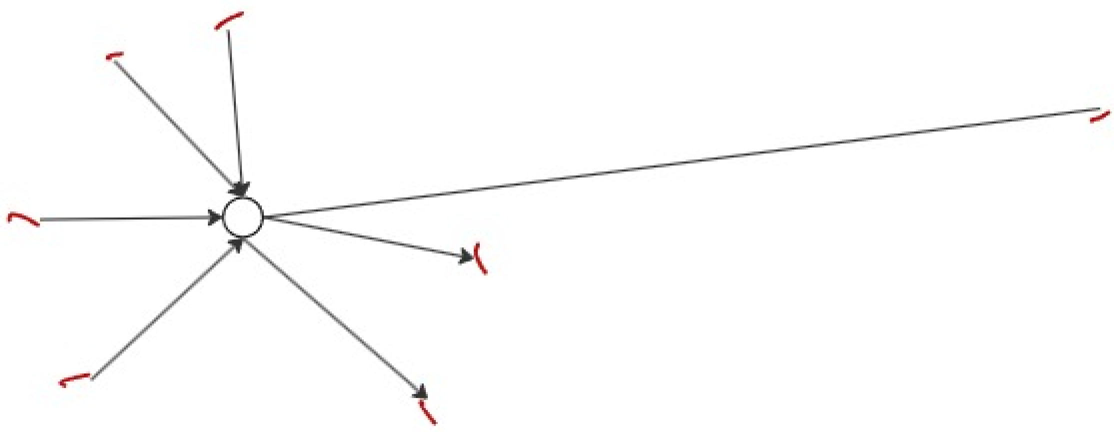


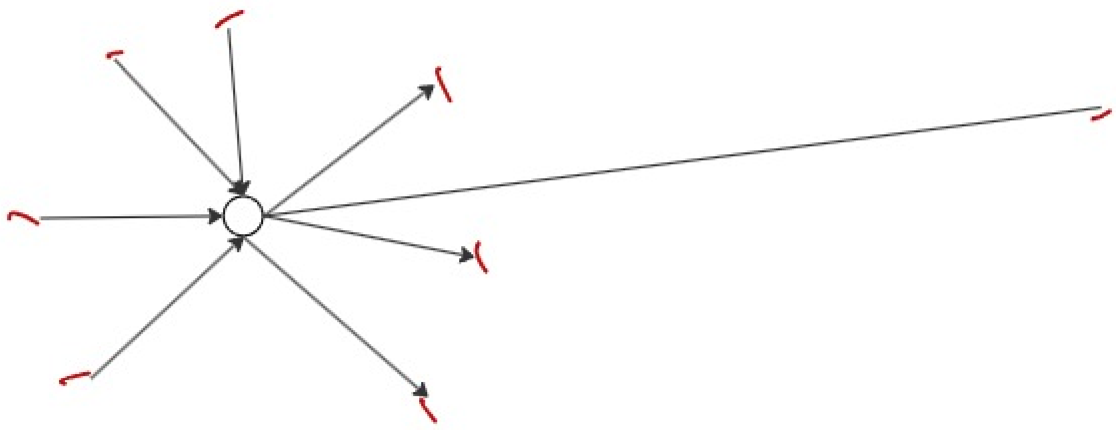
K = 1

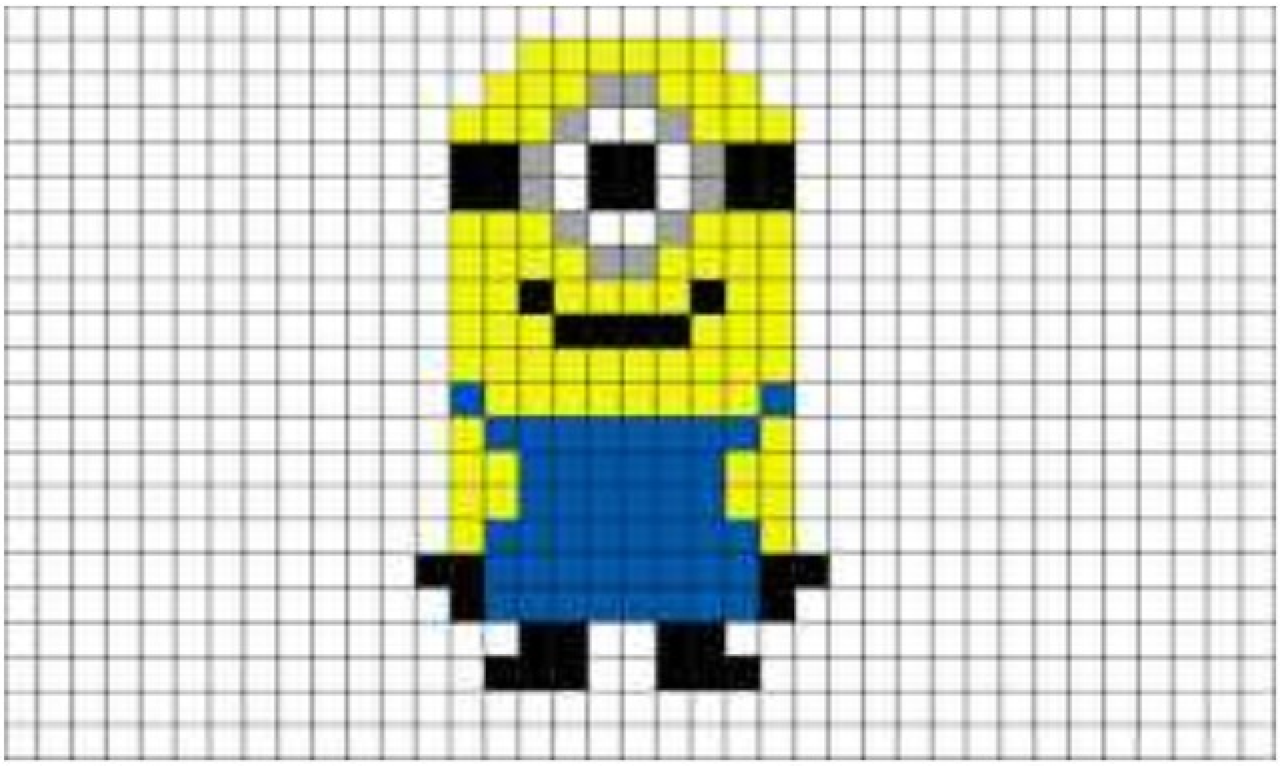
K=5

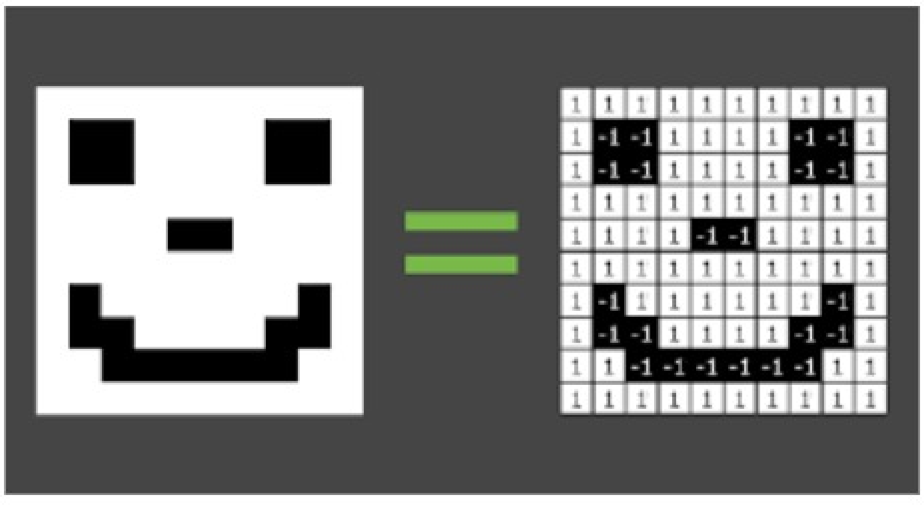
(i) (1) nn index

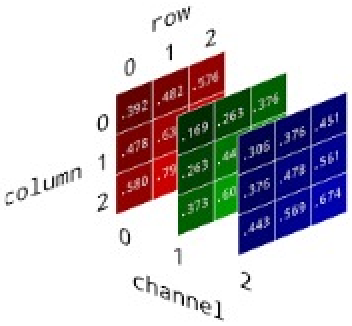
(ii) S,



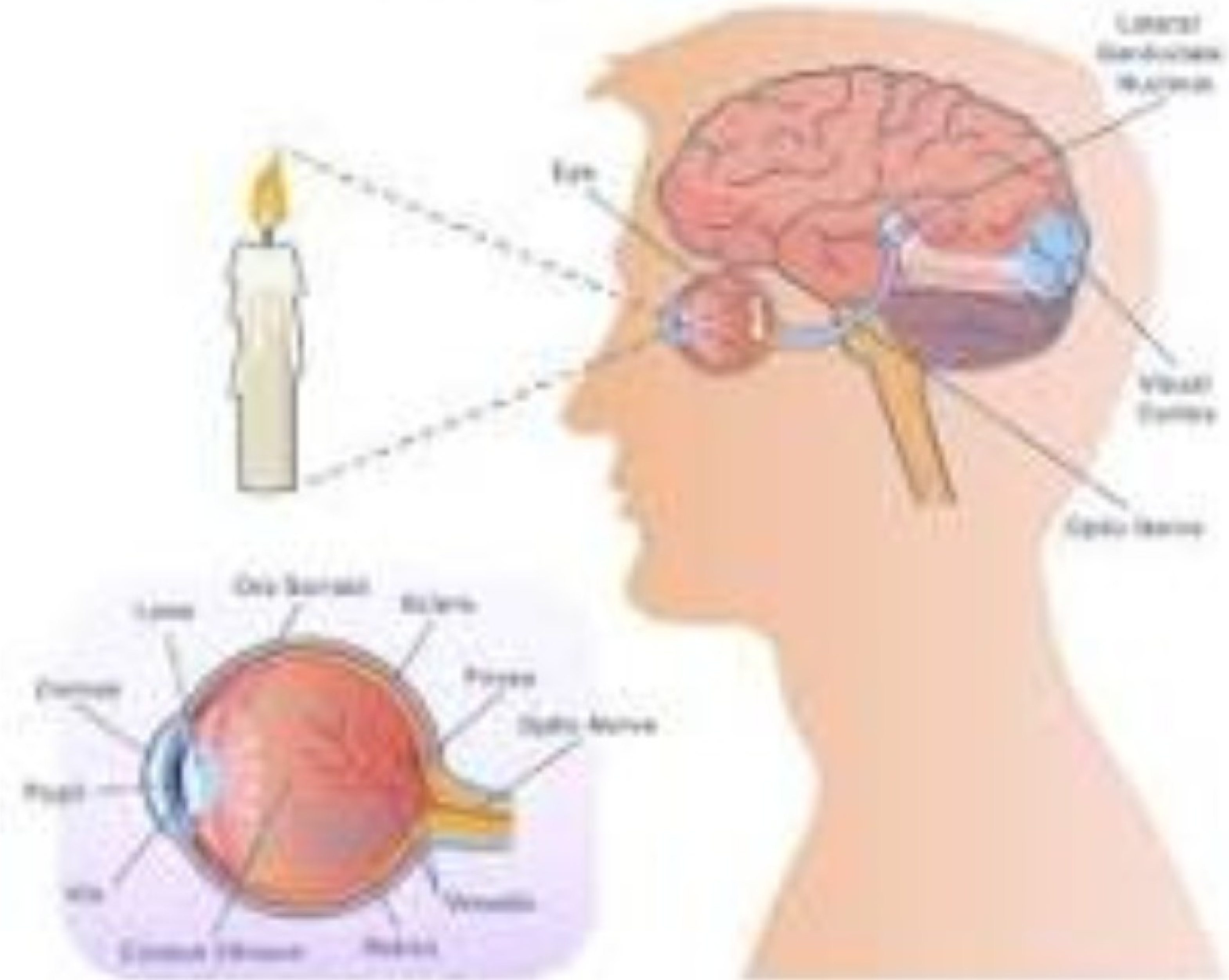


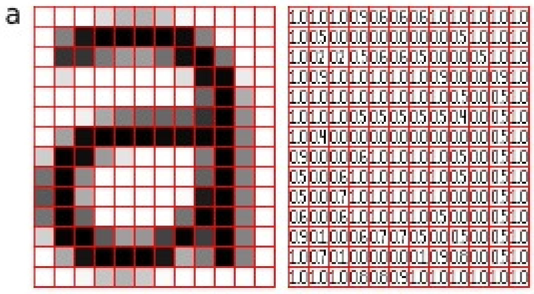


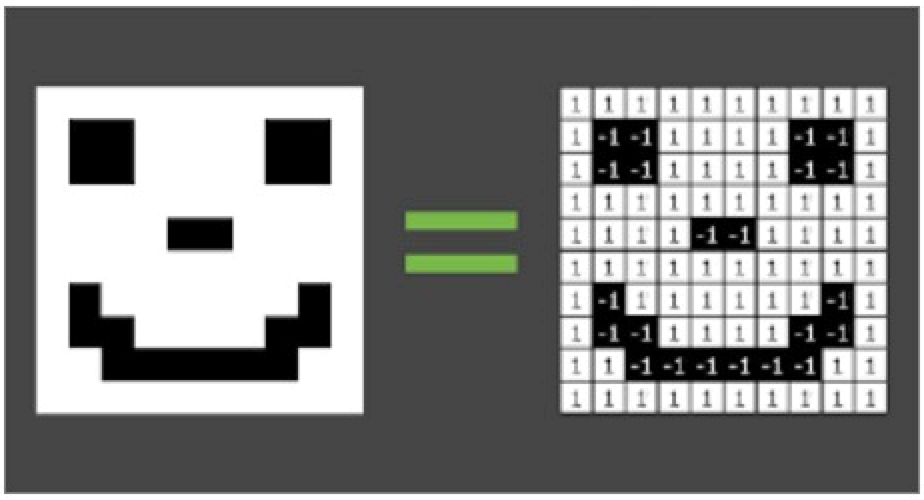

1 1 1 1 1 1 1 1 1 1
1 -1 -1 1 1 1 1 -1 -1 1
1 -1 -1 1 1 1 1 -1 -1 1
1 1 1 1 1 1 1 1 1 1
1 1 1 1 -1 -1 1 1 1 1
1 1 1 1 1 1 1 1 1 1
1 -1 1 1 1 1 1 1 -1 1
1 -1 -1 1 1 1 1 -1 -1 1
1 1 -1 -1 -1 -1 -1 -1 1 1
1 1 1 1 1 1 1 1 1 1

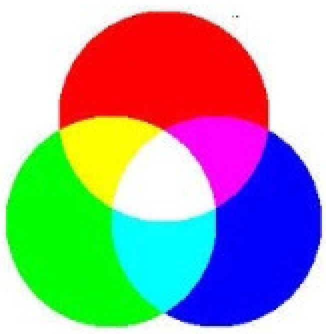

row
0 1 2
column
0 1 2
channel
0 1 2

HOW THE EYE WORK

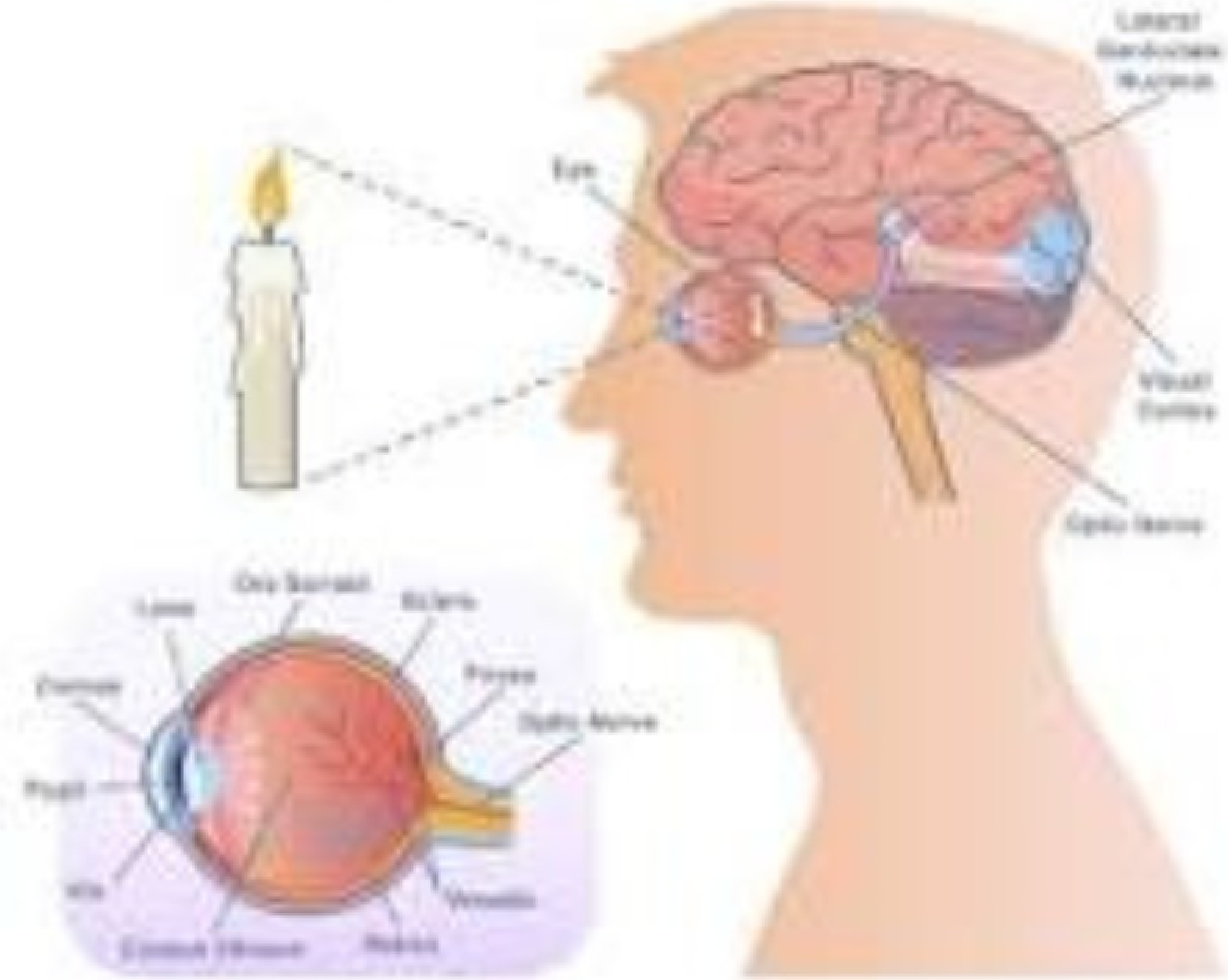


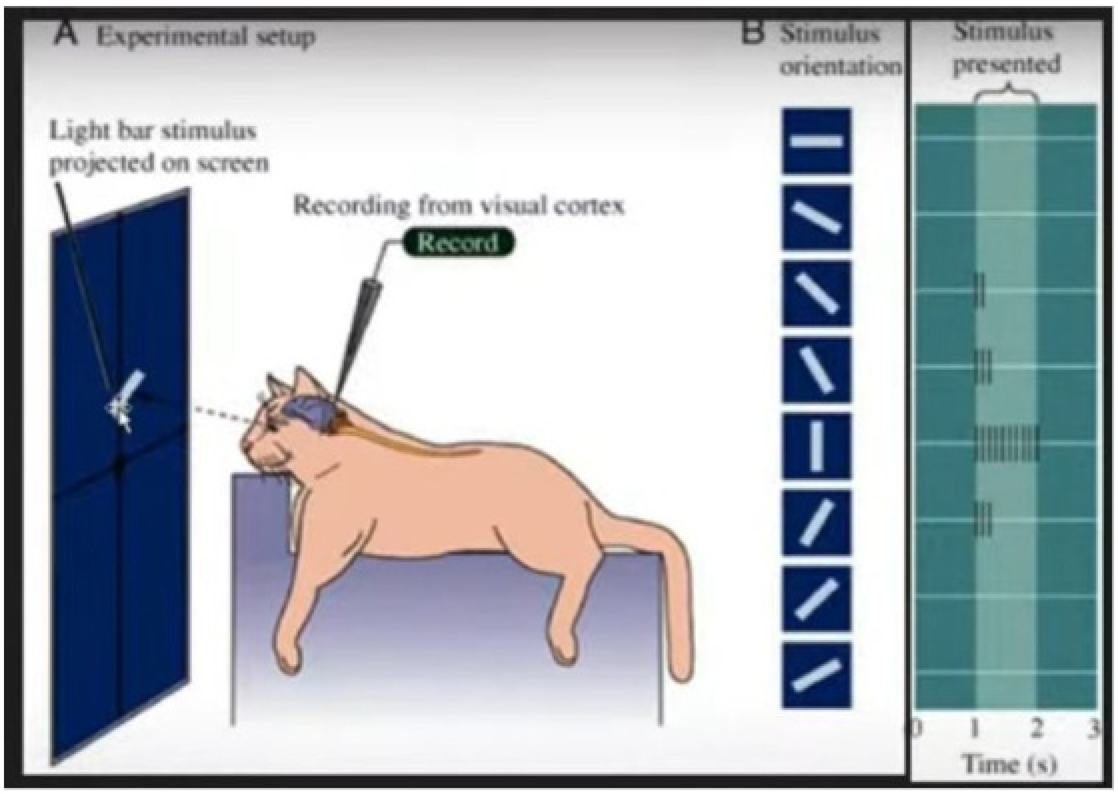

a

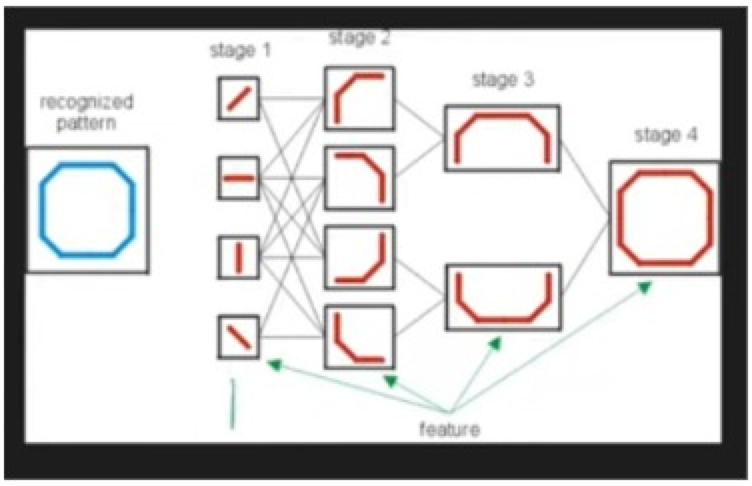

1 1 1 1 1 1 1 1 1 1
1 -1 -1 1 1 1 1 -1 -1 1
1 -1 -1 1 1 1 1 -1 -1 1
1 1 1 1 1 1 1 1 1 1
1 1 1 1 -1 -1 1 1 1 1
1 1 1 1 1 1 1 1 1 1
1 -1 1 1 1 1 1 1 -1 1
1 -1 -1 1 1 1 1 -1 -1 1
1 1 -1 -1 -1 -1 -1 -1 1 1
1 1 1 1 1 1 1 1 1 1



HOW THE EYE WORK




A Experimental setup
B Stimulus orientation
Stimulus presented
Light bar stimulus projected on screen
Recording from visual cortex
Record
0 1 2 3
Time (s)


recognized pattern
stage 1
stage 2
stage 3
stage 4
feature

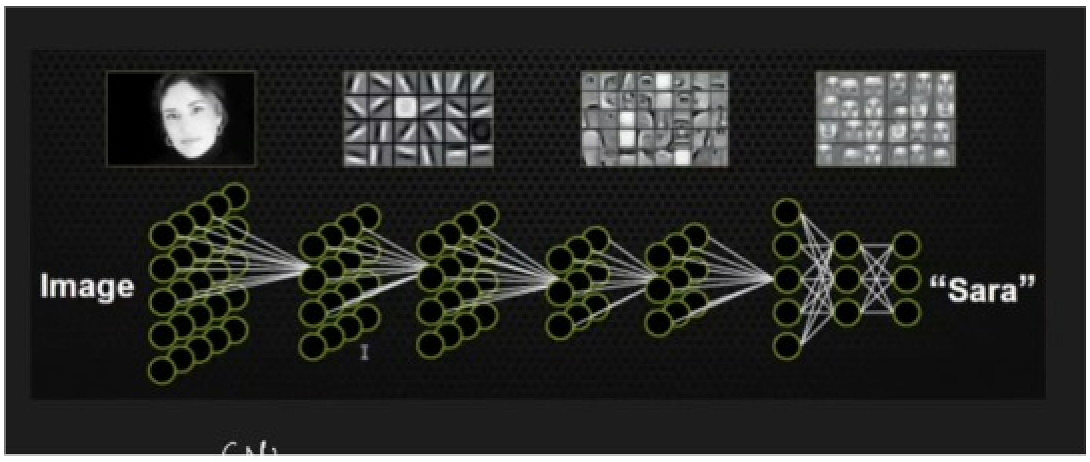
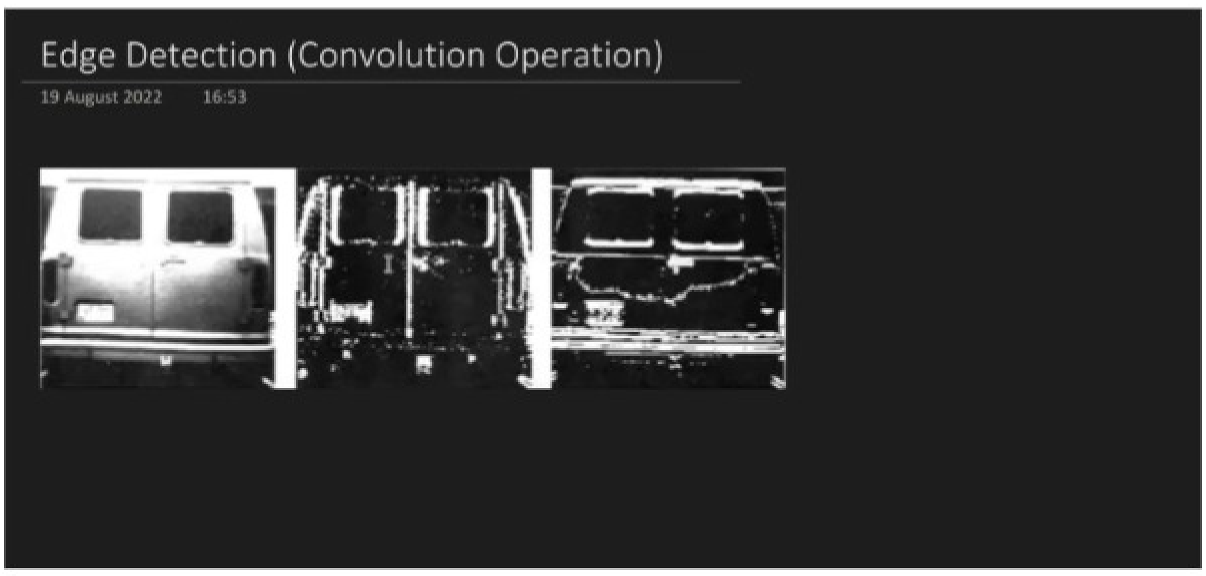
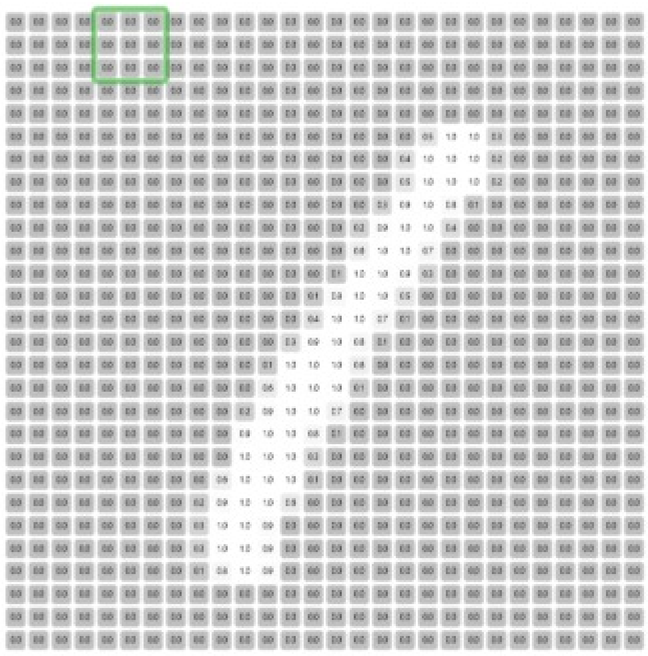


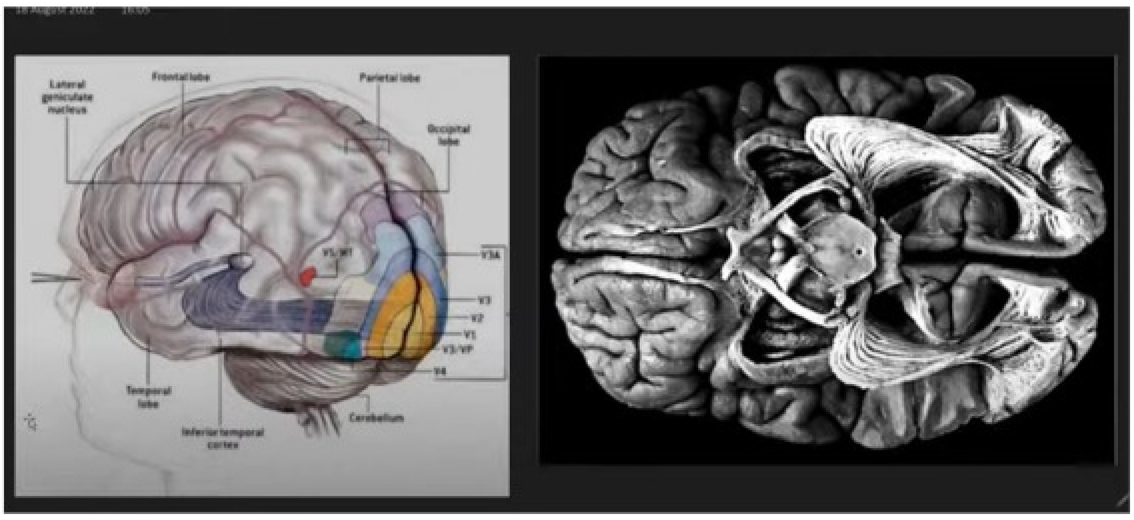
Image
“Sara”


Edge Detection (Convolution Operation)
19 August 2022 16:53

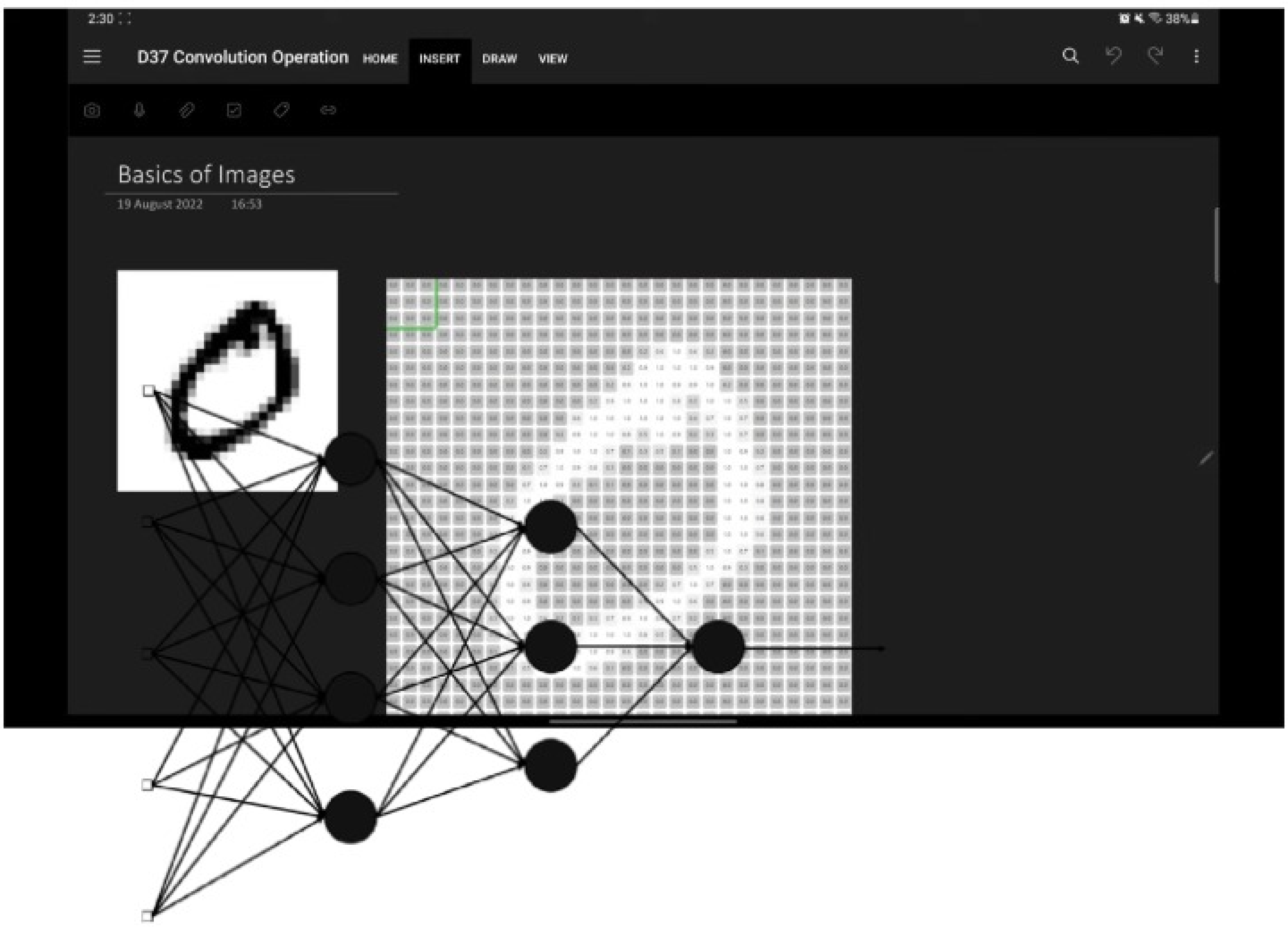


Filter

-1.0	-1.0	-1.0
1.0	1.0	1.0
0.0	0.0	0.0



https://deeplizard.com/resource/pavq7noze2



TF-IDF (Term Frequency-Inverse Document Frequency)

Suppose we have the following 2 documents:

- **Doc 1:** "I love dogs"
- **Doc 2:** "I love cats"

From both documents, we extract the unique words (just like BoW):

- I, love, dogs, cats

Term Frequency (TF)

Column 1	Column 2	Column 3
Word	TF in Doc 1	TF in Doc 2
I	1/3	1/3
love	1/3	1/3
dogs	1/3	0
cats	0	1/3

: Inverse Document Frequency (IDF)

Column 1	Column 2	Column 3
Word	Document Frequency	IDF
I	2	log(2 / (1+2)) = log(2 / 3) ≈ 0
love	2	log(2 / (1+2)) ≈ 0
dogs	1	log(2 / 1) ≈ 0.693
cats	1	log(2 / 1) ≈ 0.693

TF × IDF

Column 1	Column 2	Column 3
Word	TF-IDF in Doc 1	TF-IDF in Doc 2
I	1/3 × 0 = 0	1/3 × 0 = 0
love	1/3 × 0 = 0	1/3 × 0 = 0
dogs	1/3 × 0.693 ≈ 0.231	0 × 0.693 = 0
cats	0 × 0.693 = 0	1/3 × 0.693 ≈ 0.231

✅ Result:

- **Doc 1 vector**: [0, 0, 0.231, 0]
- **Doc 2 vector**: [0, 0, 0, 0.231]